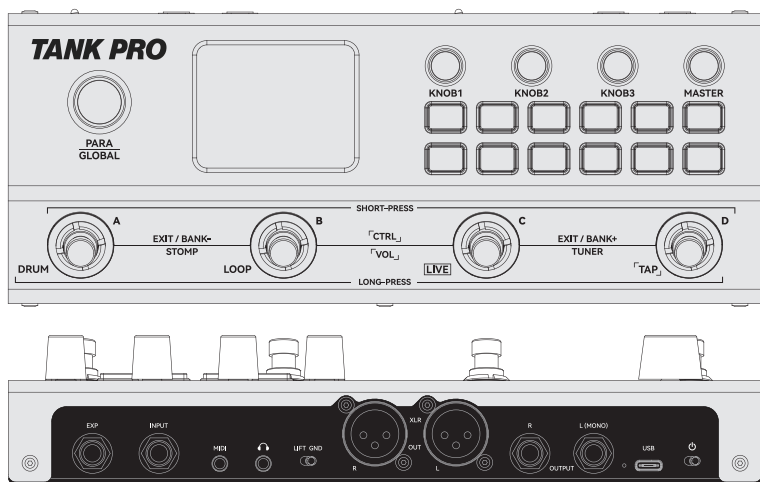


TANK PRO

A professional effects processor

User Manual | 说明书



I. Preface

TANK PRO is a professional effects processor equipped with extensive effect modules, offering flexible customization and multi-scenario adaptability. It is designed for tone shaping and performance creation with electric guitars, electric basses, and other instruments.

This manual comprehensively covers the device's safety specifications, panel operations, mode settings, effect parameters, and customization procedures, aiming to help you quickly master its core functions and unlock diverse sonic possibilities.

To ensure safe use and maximize device performance, it is recommended to read this manual carefully before operation. For in-depth understanding of specific functions, please refer to the detailed instructions in the corresponding sections.

II. Usage Instructions

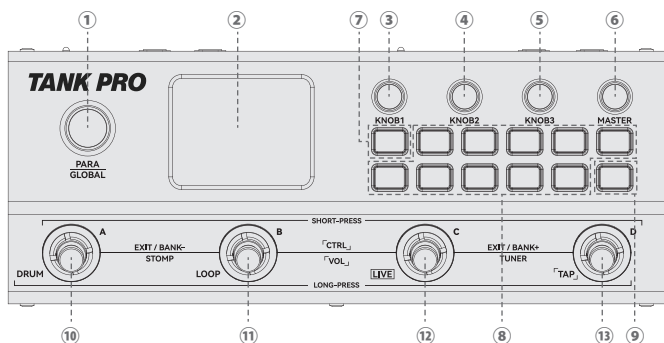
Before using this device, please read the following instructions carefully:

1. **Safety Warning:** Do not disassemble or modify the device without authorization. Unauthorized alterations may cause device damage or pose personal safety risks.
2. **Power Management:** The device is equipped with a built-in battery that supports simultaneous charging and use. When the battery is low, recharge promptly to avoid unstable device operation, it is recommended to use a 5V/2A power adapter for charging.
3. **Connection Operation:** Before connecting or disconnecting cables between this device and other equipment, ensure that all related devices are powered off.
4. **Anti-Interference Requirement:** Electromagnetic waves from equipment such as radios or televisions may interfere with this device. It is recommended to maintain sufficient distance between this device and such equipment.
5. **Placement Environment Requirements:** To prevent deformation, discoloration, or other damage, do not place the device in the following environments:
 - a. Near any heat source
 - b. Areas with strong magnetic fields
 - c. Locations exposed to direct sunlight
 - d. High-temperature or humid environments
 - e. Dusty or unclean places
 - f. Environments with strong vibration or shaking

III. Glossary

- **GLOBAL SAVE:** A function to save the device's global settings (common configurations), used for storing basic operational parameters at the overall device level.
- **PATCH SAVE:** A function to save individual presets (including effect programs, tone parameters, etc.), used for storing parameter configurations corresponding to specific tone schemes.
- **SYNC:** A parameter synchronization function among various internal modules of the device, ensuring that parameters across modules remain matched and consistent.
- **VOL Mode:** Preset volume adjustment mode, used for independently adjusting the output volume of the corresponding preset.
- **CTRL Mode:** Footswitch function customization mode, allowing personalized configuration of the device's footswitch functions according to practical usage needs.
- **TAP Mode:** Preset BPM (beats per minute) adjustment mode, used for setting or adjusting the tempo parameters of the corresponding preset.
- **LIVE Mode:** This mode is suitable for live performances. Four preset names and their numbers for the footswitches will be displayed on the screen, allowing you to check the current four preset names at any time during the performance. When switching banks, you can also view the four preset names of the target bank.
- **Short Press:** The pressing and releasing actions are completed within 1.5 seconds, it takes effect the moment it's released.
- **Long Press:** Hold down the footswitch and release it only after more than 1.5 seconds, it takes effect at this moment of 1.5s.

IV. Front Panel Overview



Main Panel Components Description

1. PARA/GLOBAL Knob (Rotary Encoder with Press Function)

- Short Press: Enter/Exit an item, execute confirmation.
- Rotate: Select items, adjust parameters, move cursor.
- Long Press: Enter GLOBAL settings interface; settings are saved automatically upon exit.

2. Display Screen: 2.4-inch LCD, displays current device operating status.

3. KNOB1 Custom Knob: Factory default for adjusting the AMP module's Gain parameter.

4. KNOB2 Custom Knob: Factory default for adjusting the AMP module's Level parameter.

5. KNOB3 Custom Knob: Factory default for adjusting the AMP module's Middle parameter.

6. MASTER Knob: Control the output volume.(Applies to the IN interface, DRUM, and LOOP, but does not apply to USB and BT. Please adjust MUSIC parameter in the HOME interface.)

* Note:

When connecting your mobile phone via BT or your computer via USB to play music, please remember to turn down the volume on your phone/computer or reduce the MUSIC parameter beforehand to avoid excessively loud sound.

7. HOME Illuminated Button/BT switch Button(with BT states light)

- Long Press: Controls BT ON/OFF (flashes when unconnected, stays lit when connected. When the light is off, it indicates that BT is turned off.).
- Short Press on Main Interface: Enters STOMP interface; press again to return to main interface.

8. Effect Module Illuminated Buttons (corresponding to WAH/FX1/FX2/DS/AMP/CAB/EQ/MOD/DLY/REV modules)

- Short Press: Enters the corresponding module interface; press again to enable/disable the module.
- Long Press: Toggles the module ON/OFF.
- When module is ON: Button remains lit.

9. SAVE Illuminated Button

- Short Press: Enters the preset save interface; press again within the interface to execute PATCH SAVE.
- After preset parameters are modified: The button flashes automatically. Therefore, the SAVE button light blinks to indicate whether the current default parameter has been changed, if the light blinks, it indicates that the parameter has been changed. Before you switch the preset, you'd better save the parameter in time to avoid data loss.
- Long Press: Executes GLOBAL SAVE.

10. A Illuminated Footswitch: Press and hold to enter the DRUM interface; press and hold again to return to the main interface.

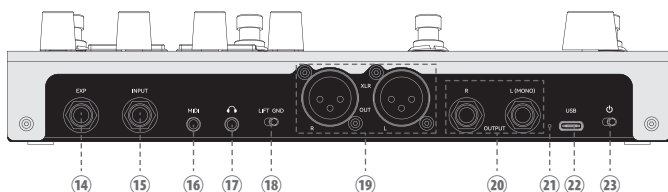
11. B Illuminated Footswitch: Press and hold to enter the LOOP interface; press and hold again to return to the main interface.

12. C Illuminated Footswitch: Press and hold to enter LIVE mode; press and hold again to return to the main interface.

13. D Illuminated Footswitch: Press and hold to enter TAP mode; press and hold again to return to the main interface.

* Combination Footswitch Operations

- 1. Press and Hold A+B:** Enter the STOMP interface; press and hold again to return to the main interface.
- 2. Press and Hold C+D:** Enter the TUNER interface; press and hold again to return to the main interface.
- 3. Press and Hold B+C:** Enter Preset VOL Mode.
 - In this mode: Short/Long press B to decrease volume; Short/Long press C to increase volume.
 - Exit method: Long press B+C.
- 4. Short Press B+C:** Enter the CTRL footswitch customization interface; press again to return to the previous interface.
- 5. Short Press A+B or C+D** (When not on the main interface and LIVE interface): Return to the main interface.
- 6. Short Press A+B or C+D** (When on the main interface and LIVE interface): Used for preset bank switching.



Connections and Auxiliary Components Description

14. EXP Jack: 6.35mm (1/4") TRS jack for connecting an external expression pedal (Notes: please use 1/4" TRS cable). The pedal can be set to P1/P2 states, you can adjust two parameters of two different states with one pedal. The parameters of P1/P2 can be customized, and the P1/P2 states can be changed by the footswitches on the CTRL interface.

15. INPUT Jack: 6.35mm (1/4") mono input jack for connecting electric guitars, electric basses, and other instruments.

16. MIDI Jack: 3.5mm (1/8") TRS jack for connecting a MIDI controller.

17. Headphone Jack: 3.5mm (1/8") stereo output jack for connecting headphones.

18. LIFT/GND Switch

For XLR OUT Ground Lift/Ground Switch: Switch to "LIFT" for ground lift, switch to "GND" for ground connection.

19. XLR OUT Jack: XLR balanced output jack. Mono output can be connected to the left channel (L) for interfacing with mixers, audio interfaces, etc.

20. OUTPUT Jack: 6.35mm (1/4") TS unbalanced output jack. Mono output can be connected to the left channel (L/MONO) for interfacing with instrument amplifiers, etc.

21. Charging Indicator: Off during normal operation when powered on. Lights red while the device is charging. Turns blue when fully charged.

22. USB Port (USB TYPE-C)

● Basic Functions: Charges the device battery; connects to a host computer (enabling parameter editing, preset backup, DS/AMP/CAB tone import, firmware updates, factory reset).

● Advanced Functions: Supports USB MIDI, USB AUDIO, computer audio interface functionality, and computer data transfer, as well as OTG internal recording.

● Note: Using the included USB cable with a separately purchased OTG adapter allows connection to third-party mobile apps for synchronized audio-video internal recording.

23. Power Switch: Controls turning the device on and off.

1. KNOB Customization

Mapping Steps

- ① After entering an effect module interface, select the target control parameter using the PARA knob, then press the PARA knob briefly.
- ② When the parameter selection box changes color, turn the KNOB; once the pop-up prompt appears, the mapping is complete.
- ③ Press the PARA knob briefly again to exit the selection state. The KNOB can now provide custom control for that parameter.

2. CTRL Footswitch Customization

Interface Description

The top four function slots on the CTRL interface correspond to the short-press functions of footswitches A, B, C, and D. The bottom four slots correspond to their long-press functions. (For the sake of convenience and simplicity, only the function of a single footswitch can be customized at preset. Because the function of combination footswitches is already defined by the designer, do not support modification.)

Mapping Steps

- ① Enter the CTRL interface, use the PARA knob to select the target footswitch.
- ② Press the PARA knob briefly to enter the function selection interface, then select the function to be mapped.
- ③ Press the PARA knob briefly to confirm. The device automatically returns to the CTRL interface, indicating the mapping is complete.

Footswitch Mapping Function Table

Short Press	Long Press	Description
MODE CTRL	MODE CTRL	Multi-Module Switch
PEDAL SWITCH	PEDAL SWITCH	EXP External Expression Pedal State Switch (P1/P2)
DLY FREEZE	DLY FREEZE	Delay Effect Freeze (The freeze effect can only be maintained in the CTRL interface. Once the preset is changed or the CTRL interface is exited, the freeze effect will immediately become invalid.)
KILL SWITCH	KILL SWITCH	Mute (Multiple Short Presses: Instant Mute; Long Press: Mute, Release to Restore) — Mapping occupies both top and bottom slots
PRESET+1	PRESET+1	Preset +1
PRESET-1	PRESET-1	Preset -1
PRESET+4	PRESET+4	Preset +4 (Preset Bank Jump)
PRESET-4	PRESET-4	Preset -4 (Preset Bank Jump)
VOL+5	VOL+5	Preset Volume +5
VOL-5	VOL-5	Preset Volume -5
BPM+1	BPM+1	Preset BPM +1
BPM-1	BPM-1	Preset BPM -1
MOD TAP	-	Adjust MOD Module SPEED Parameter (Tap Tempo via consecutive short presses)
DLY TAP	-	Adjust DLY Module TIME Parameter (Tap Tempo via consecutive short presses)
DRUM TAP	-	Adjust DRUM BPM (Tap Tempo via consecutive short presses)
BPM TAP	-	Adjust Preset BPM (Tap Tempo via consecutive short presses)
LOOP	LOOP	Start LOOP recording or playback
DRUM	DRUM	DRUM On/Off
DRUM+1	DRUM+1	DRUM Pattern +1
DRUM-1	DRUM-1	DRUM Pattern -1
BYPASS	BYPASS	Bypass Mode
MUTE	MUTE	Mute Mode
PATCH SAVE	PATCH SAVE	Save Preset Settings
LOOP STOP	LOOP STOP	STOP LOOP
NONE	NONE	Clear Footswitch Mapping Function

3. EXP External Expression Pedal Customization

Prerequisites

To use an external pedal for controlling module parameters, after connecting the pedal, enter the PEDAL interface under GLOBAL to complete pedal calibration, once calibration is complete, proceed with the following mapping steps.

Mapping Steps

- ① Enter an effect module interface, select the parameter to be controlled using the PARA knob, then briefly press the PARA knob.
- ② After the parameter selection box changes color, depress the external pedal. The pop-up prompt indicates that mapping is complete.
- ③ Briefly press the PARA knob again to exit the selection state. The EXP external expression pedal can now provide custom control for that parameter.

P1/P2 State Switching and Saving

- To set P2 after configuring P1, enter the CTRL interface or enter "GLOBAL-PEDAL-SWITCH" to switch the pedal state, then repeat the mapping steps above.
- The pedal's P1 and P2 states are saved with the preset: when the pedal is in P1 state for the current preset, saving the preset will cause the pedal to default to P1 state the next time this preset is recalled.

4. Note

The aforementioned custom settings for KNOBs, CTRL footswitches, and external expression pedals are all bound to and saved with their corresponding presets. When performing a PATCH SAVE operation, such custom settings are stored simultaneously with other parameters of that preset. When switching presets, the corresponding preset's custom settings will also take effect accordingly.

VI. Mode & Interface Introduction and Operation Logic

*** Tips:**

Note on the design philosophy of entering and exiting various operation interfaces using footswitches on the main screen:

Our design works as follows: after entering a specific interface by long-pressing or short-pressing one or two footswitches, you can exit that interface by performing the same action.

Combination Footswitch Operations

PRESET -1	PRESET +1	VOL -5	VOL +5	← Footswitch Short-Press Functions
PRESET -4	PRESET +4	PEDAL SWITCH	PATCH SAVE	

- 1.Entry Method:** In any interface, perform a short press on the B&C footswitch combination to enter CTRL mode.
- 2.Core Feature:** The footswitch customization function is active only in CTRL mode (factory default settings are shown in the above picture).
- 3.Footswitch Function Distribution**

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	Customize	Customize	Customize	Customize	Exit current interface	Exit current interface	Exit current interface
Long Press	Customize	Customize	Customize	Customize	-	-	-

VOL Mode

VOL

60

1.Entry Method: In any interface (outside CTRL modes), perform a long press on the B&C footswitch combination to enter VOL mode.

2.Core Feature: VOL mode is a preset-dedicated volume adjustment mode. It supports "fixed-step volume adjustment" and "continuous volume adjustment" via designated footswitches, providing a specialized function for rapid, precise preset volume control in performance scenarios.

3.Footswitch Function Distribution:

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	-	Single-Step Volume Decrease	Single-Step Volume Decrease	-	Exit current interface	Enter CTRL mode	Exit current interface
Long Press	-	Continuous Volume Decrease	Continuous Volume Decrease	Enter TAP mode	-	Exit current interface	-

4.Mode Indication

When VOL mode is activated, only indicators for B and C pedals remain illuminated, providing clear visual confirmation of the current volume adjustment status.

TAP Mode

BPM

80

1.Entry Method: In any interface (outside CTRL modes), perform a long press on footswitch D to enter TAP mode.

2.Core Feature: TAP mode is a preset-specific BPM adjustment function mode that supports "tap tempo" operation, allowing for quick synchronization with the performer's real-time rhythm. If the SYNC function is enabled in the corresponding effect module, the module will operate in sync with the BPM set in TAP mode, making it a core feature for synchronizing song tempo during live performances and creative processes.

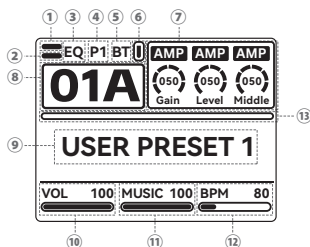
3.Footswitch Function Distribution:

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	-	Single-Step BPM Decrease	Single-Step BPM Decrease	Tap Tempo via Consecutive Short Presses (Set BPM)	Exit current interface	Enter CTRL mode	Exit current interface
Long Press	-	Continuous BPM Decrease	Continuous BPM Decrease	Exit TAP Mode	-	Enter VOL mode	-

4.Mode Notes:

● After entering TAP mode, the LED indicator on footswitch D flashes in sync with the current BPM, providing intuitive feedback on the tempo.

Main Interface



I. Display Content

1.INPUT Signal Indicator: Shows the real-time audio signal status being received at the INPUT port.

2.Mixed Output Signal Display: Shows the real-time audio signal status at the device's mixed output.

3.Global EQ Status Display: Remains constantly lit when activated, the lights will go out when deactivated.

4.EXP Pedal Status Indicator: Displays the operating mode (P1/P2) of the external expression pedal; status changes require PATCH SAVE to store.

5.BT Status Indicator: BT status is saved independently and autosaves upon changes.

- Connected: Steady light.
- ON but unconnected: Flashing.
- OFF: No light.

(The BT status is synchronized with the HOME button flashing status.)

6.Battery Level Indicator.

7.Custom Knob Control Parameter Display (KNOB1/KNOB2/KNOB3):

- Factory Default: Corresponds to AMP module Gain, Level, and Middle parameters respectively.
- Parameter mapping changes require PATCH SAVE to store.

8.Preset Position Display:

- Grouping Logic: Organized by BANK groups (01 being the first group), with A, B, C, D corresponding to 4 tones per group.
- Storage Capacity: 30 groups totaling 120 preset slots.
- Factory Allocation: Groups 01-15 (Factory Sounds), Groups 16-30 (no-effect Sounds).

* Tips:

After you switch the preset, the preset position will not be automatically saved. If you need to store the position, before shutting down the device, you need to long-press the SAVE button manually. Then, the screen will pop up "Global save", and you can get the preset position before shutting down the device in the next startup.

9.Preset Name Display: Can be modified in the SAVE interface; modifications require PATCH SAVE to store.

10.VOL (Preset Volume) Display: Displays current preset volume parameter; changes require PATCH SAVE to store.

11.MUSIC Volume Display: Displays BT and USB input audio source volume parameters; changes require GLOBAL SAVE to store.

12.BPM (Preset Tempo) Display: Displays current preset BPM parameter; changes require PATCH SAVE to store.

13.Looper display bar: If there is audio in the Looper, this display bar will be displayed on the main screen.

* Relationship Between Preset BPM and Effect SYNC Switch

Core Function: Controls the linkage between parameters (Time/Speed/DRUM BPM) and the preset BPM, enabling synchronization or independent control of effect speed with the musical tempo.

* SYNC OFF State

- Parameter Units: Time (delay time, tremolo period, etc.), Speed (filter sweep rate, LFO rate, etc.) are set in absolute time units (e.g., ms).
- Control Method: Manual parameter adjustment, independent of preset BPM.
- Effect Characteristic: Effect speed is fixed and unaffected by changes in preset BPM.

* SYNC ON State

- Parameter Units: Time, Speed switch to relative beat units (e.g., 1/4 note, 1/8 note, including dotted/triplet subdivisions).
- Control Method: Actual parameter speed is automatically calculated based on the preset BPM.
- Effect Characteristic: Effect speed synchronizes with the musical tempo and automatically adjusts when the preset BPM changes.

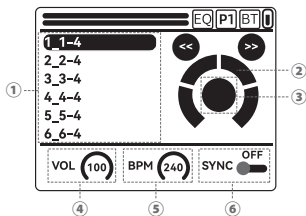
II. Footswitch Functions

On the main interface, the long/short press functions of individual and combination footswitches are fixed, with specific distribution as follows:

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	BANK (Preset A in Group)	BANK (Preset B in Group)	BANK (Preset C in Group)	BANK (Preset D in Group)	BANK- (Group -1)	Enter CTRL Mode	BANK+ (Group +1)
Long Press	Enter DRUM Interface	Enter LOOP Interface	Enter LIVE Mode Interface	Enter TAP Mode	Enter STOMP Interface	Interface Enter VOL Mode	Enter TUNER Interface

DRUM Interface

Entry Method: In any interface (outside CTRL modes), perform a long press on footswitch A to enter DRUM mode.



I. Interface Display Content

- 1.Drum Machine Pattern List:** Displays selectable drum machine rhythm patterns (100 total).
- 2.Beat Count Display:** Shows the current running beat count of the drum machine.
- 3.First Beat Indicator:** Turns green on the first beat to assist with rhythm alignment.
- 4.DRUM Volume:** Adjusts the drum machine output volume.
- 5.DRUM BPM.**
- 6.Sync Switch:** Controls whether the drum machine BPM synchronizes with the preset BPM.

* Note:

Settings within the DRUM interface can be saved via GLOBAL SAVE.

II. Footswitch Functions

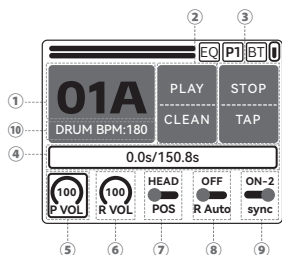
Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	DRUM Pattern -1	DRUM Pattern +1	DRUM On/Off	Adjust DRUM BPM (Tap Tempo via consecutive short presses)	Exit DRUM Interface	Enter CTRL Mode Interface	Exit DRUM Interface
Long Press	Exit DRUM Interface	Enter LOOP Interface	Enter LIVE Mode Interface	Enter TAP Mode	Enter STOMP Interface	Enter VOL Mode	Enter TUNER Interface

III. Footswitch LED Indicator Rules

- When the drum machine is ON, footswitch C LED remains steadily lit; when OFF, the LED is off.
- While in the drum machine DRUM interface, footswitch D LED flashes in sync with the drum machine BPM, providing intuitive tempo feedback.

LOOP Interface

Entry Method: In any interface (outside CTRL modes), perform a long press on footswitch B to enter LOOP mode.



I. Display Content

1.Preset Number Display: Shows the current preset number.

2.C Footswitch Function Display: Top indicates short-press function, bottom indicates long-press function.

3.D Footswitch Function Display: Top indicates short-press function, bottom indicates long-press function.

①REC: Start recording, REC will be displayed when there is no music in the LOOPER.

②PLAY: When the music stops, click this footswitch to play music; When recording in double overlay, click this footswitch to stop recording and play music.

③CLEAN: When the music stops, or while the first layer audio is being recorded, hold this Footswitch to delete the music, empty the LOOPER's music.

④STOP: Click the fourth footswitch to stop the recording and save, or stop playing.

⑤TAP: Hold the fourth footswitch to enter the TAP interface.

⑥BACK: It can undo/redo last layer audio unlimited times.

⑦DUB: Double, this function support unlimited overdub.

4.LOOP Duration Display: Displays total loop duration (maximum support 150.8 seconds).

5.LOOP Playback Volume: Adjusts output volume during loop playback.

6.LOOP Recording Volume: Adjusts input volume during loop recording.

7.LOOP Position in Effect Chain: Selects LOOP position in the effect chain (HEAD/TAIL).

8.LOOP Auto-Recording Sensitivity Level: Four levels: OFF, HIGH, MID, LOW; With this function enabled, the LOOP instantly starts recording as soon as you play, allowing for AUTO REC without foot control. A HIGH setting means high sensitivity, where even a light touch triggers recording. Conversely, a LOW setting indicates lower sensitivity, requiring you to play with more force to activate it.

* Note:

The R Auto function is only available in the LOOP interface.

9.DRUM SYNC: After opening, the drum will start simultaneously with LOOP, and the drum and LOOP will be automatically aligned the timeline. The difference between the functions of ON1 and ON2 is that the DRUM in ON2 will pause with the LOOP, while the DRUM in ON1 will not pause with the LOOP.

If you need to use this function, please turn it on before recording the first layer of audio, and you can achieve perfect alignment of the drums and the audio. If you record the audio first and then turn on this function, it may lead to alignment failure.

*** Note:**

Settings within the LOOP interface can be saved via GLOBAL SAVE.

10. When the DRUM SYNC function is enabled, the tempo of the drum machine will be displayed here.

II. Interface Switch Functions

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	Preset - 1	Preset + 1	Real-time changes apply. See table below.	Stop LOOP	Exit LOOP Interface	Enter CTRL Mode Interface	Exit LOOP Interface
Long Press	Enter DRUM Interface	Exit LOOP Interface	Real-time changes apply. See table below.	Enter TAP Mode	Enter STOMP Interface	Enter VOL Mode	Enter TUNER Interface

*** Note:**

When in the Loop interface, it is recommended not to use the combination of the C footswitch with other footswitches (such as B&C and C&D as shown in the diagram above). This is because the C footswitch in the Loop interface takes effect immediately when pressed, so using footswitch combinations may cause functional conflicts.

C Switch Operation Logic	
Operation Scenario	Functional Effect
If the LOOP has audio and is paused, single-press the pedal.	Play LOOP Audio
If the LOOP is empty, single-press the pedal	Initiate LOOP Recording
Single Press After Recording Completion	Initiate LOOP Playback
Single Press During Playback State	Overlay Audio Track
Long Press During Recording / Playback	Delete / Undo / Restore Audio Track ("Delete" for single track; "Undo / Restore Last Recording" for overlaid tracks)
Double Press During Recording / Playback	Pause LOOP (Recording / Playback Temporarily Stops)
Long Press After Pausing	Delete All Recording Data Under Current Preset

III. Footswitch Indicator Rules

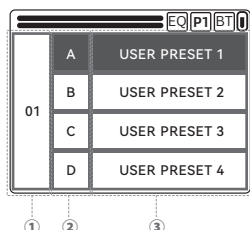
- When the LOOP recording track is empty, the C footswitch light remains off;
- When the recording track contains content, the C footswitch light illuminates; during basic LOOP operations, the light color synchronizes with the color of the C footswitch function display area.

A Footswitch

- When the drum machine is activated, the A footswitch light flashes in sync with the currently set BPM rhythm;
- If the drum machine is deactivated, the A footswitch light remains steadily illuminated.

LIVE Mode Interface

Entry Method: In any interface (outside CTRL and LOOP modes), perform a long press on footswitch C to enter LIVE mode.



I. Interface Display

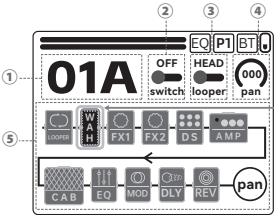
- 1.Preset BANK Number: Displays the number of the BANK to which the current preset belongs.
- 2.Corresponding Footswitch for Preset: Identifies the footswitch associated with the current preset.
- 3.Preset Name: Shows the custom name of the current preset.

II. Footswitch Functions

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	BANK (Preset A in Group)	BANK (Preset B in Group)	BANK (Preset C in Group)	BANK (Preset D in Group)	BANK- (Group -1)	Enter CTRL Mode Interface	BANK+ (Group +1)
Long Press	Enter DRUM Interface	Enter LOOP Interface	Exit LIVE Mode Interface	Enter TAP Mode	Enter STOMP Interface	Enter VOL Mode	Enter TUNER Interface

STOMP Interface

Entry Method: In any interface (outside CTRL, VOL and TAP modes), perform a long press on footswitch A+B to enter STOMP mode.



*** Convenient operation tips:**
After you save the current preset, the position of the white box will be saved with the preset. Therefore, users can enter Stomp mode by pressing the left two footswitches and then toggle a specific module on/off by click the third footswitch, without needing to pre-configure the footswitch function in the CTRL interface.

I. Interface Display

- 1.Preset Number.
- 2.Module On/Off Position: Controlled by KNOB1, adjusts the on/off status of the module.
- 3.LOOP Effect Chain Position Setting: Controlled by KNOB2, selects the position of LOOP in the effect chain (HEAD/TAIL).
- 4.Left/Right Channel Signal Output Ratio: Controlled by KNOB3, adjusts the output proportion of left and right channels.
- 5.Effect Chain Module Order: Use PARA knob to select the module and adjust its sequence in the effect chain.

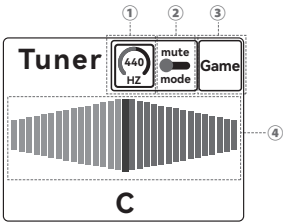
*** Note:**
1.The pan parameter, white box cursor position, and the sequence of effect modules within the STOMP interface can be saved via PATCH SAVE.
2.The Exp_Pan in the FX module is a stereo effect, belonging to the same category as the stereo effects in the MOD (Modulation), DLY (Delay), and REV (Reverb) modules. The core of these effects is based on the phase difference and time difference of dual-channel signals to create a sense of space, while mono effects independently process single-channel signals only. There is an essential difference in the signal processing logic between the two. Therefore, all stereo effects (including Exp_Pan and the stereo effects in MOD/DLY/REV) must be placed at the end of the effects chain (the final processing stage before signal output). If they are connected earlier, subsequent mono effects may disrupt the spatial phase relationship of the dual-channel signals, leading to issues such as muddy sound, positioning shifts, or a loss of spatial sense.

II. Footswitch Functions

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	Preset -1	Preset +1	Module On/Off	Cursor Cycles to Select Module (Cursor position saved per preset)	Exit STOMP Interface	Enter CTRL Mode Interface	Exit STOMP Interface
Long Press	Enter DRUM Interface	Enter LOOP Interface	Enter LIVE Mode Interface	Enter TAP Mode	Exit STOMP Interface	Enter VOL Mode	Enter TUNER Interface

TUNER Interface (GAME Mode)

Entry Method: In any interface (outside CTRL, VOL and TAP modes), perform a long press on footswitch C+D to enter TUNER mode.



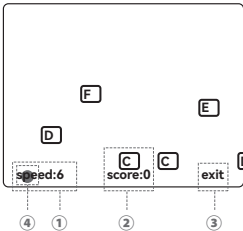
I. Interface Display

- 1.Tuning Reference Frequency Setting:** Adjusted by KNOB1 to set the tuning reference frequency (default 440Hz, corresponding to the standard pitch A₄), providing a calibration reference for instrument tuning.
- 2.MUTE/BYPASS Setting:** Controlled by KNOB2 to switch between "Mute (MUTE)" or "Bypass (BYPASS)" operation modes.
- 3.Game Switch:** Controlled by KNOB3 to turn the GAME function on or off.
- 4.Tuner Display:** Visually indicates pitch deviation via green/red bars (left shift = pitch too low, right shift = pitch too high), with specific note names displayed below to assist precise tuning.

II. Footswitch Functions

Pedal/Operation	A	B	C	D	A&B	B&C	C&D
Short Press	Mute Mode	Bypass Mode	Exit TUNER Interface	Exit TUNER Interface	Exit TUNER Interface	Enter CTRL Mode Interface	Exit TUNER Interface
Long Press	Enter DRUM Interface	Enter LOOP Interface	Enter LIVE Mode Interface	Enter TAP Mode	Enter STOMP Interface	Enter VOL Mode	Exit TUNER Interface

III. GAME Mode Interface



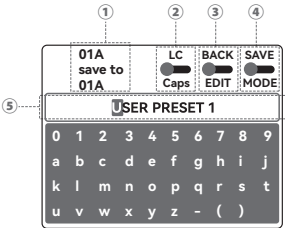
Interface Display

- 1.Note Scrolling Speed Adjustment:** Controlled by KNOB1 to adjust the scrolling speed of note names on the interface.
- 2.Score Management:** Displays real-time GAME score on the interface; current score can be cleared via KNOB2.
- 3.Game Exit Operation:** Rotate KNOB3 to exit the GAME interface.
- 4.Recognition Point Feedback:** Moves vertically on the interface based on the pitch of the input signal from the INPUT port, providing intuitive feedback on pitch recognition status.

Return Logic

In the GAME interface, briefly pressing any single footswitch (A/B/C/D) directly returns to the main TUNER tuning interface.
Briefly pressing A&B or C&D footswitches directly returns to the main home interface.

SAVE Interface



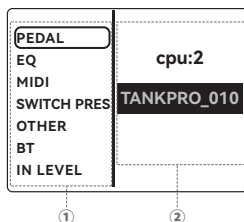
I. Interface Display and Control Elements

- 1.Preset Number and Operation Display:** Intuitively displays the current preset number, simultaneously showing two operation options: SAVE and SWAP.
- 2.Letter Case Toggle:** Controlled by KNOB1, allowing switching between uppercase and lowercase input letters.
- 3.Delete / Space Control:** Controlled by KNOB2; rotating clockwise triggers space input, while rotating counterclockwise triggers character deletion.
- 4.Operation Mode Toggle:** Controlled by KNOB3, used to switch between executing SAVE or SWAP operations.
- 5.Preset Name Editing Area:** Use the PARA knob to select individual characters within the name, then enter the keyboard interface to choose the target character for editing.

II. Save / Swap Operation Steps

- 1.After completing the preset name editing, use KNOB3 to select the desired operation (SAVE or SWAP);
- 2.Adjust and confirm the save/swap location;
- 3.Briefly press the SAVE button to confirm and execute the operation;
- 4.Upon successful operation, the interface will display a "Save OK" prompt, completing the entire process.

GLOBAL Interface



Interface Display Elements

1.Settings Menu List (left side): Displays configuration options such as PEDAL, EQ, MIDI, SWITCH PRESET, OTHER, BT and INPUT LEVEL, allowing users to select corresponding functions to access detailed configuration pages.

2.System Information Display (right side): Shows CPU operation information and displays the device firmware version identifier, facilitating easy monitoring of device status and version details. Warning, when the load rate exceeds 90 (90%), it may cause noise or other problems.

* Tips:

As soon as you exit the GLOB Interface, the system will perform a global parameter save operation. This function is the same as long pressing the SAVE button. Then, the message "Global save" will be displayed.

PEDAL (External Expression Pedal) Interface

Core Function

Used for the calibration and functional parameter configuration of an external expression pedal. Before using an external pedal, you must enter this interface to complete the setup.

I. Pedal Calibration Steps

- 1.After entering the PEDAL interface, follow the on-screen diagram prompts to complete calibration operations, such as full pedal travel triggering.
- 2.After completing each calibration step, confirm by short pressing the PARA knob or short pressing the C footswitch until the entire calibration process is finished.

II. Parameter Control (Corresponding Knob Adjustments)

1. Invert (Pedal Polarity Position)

- Control Knob: KNOB1
- Function: Switches the control direction of the pedal (Forward / Reverse) to accommodate different usage habits.

2. Toe (Module On/Off Control when the pedal switches between P1/P2 states)

- Control Knob: KNOB2

Position Description:

- None: When the pedal switches between P1/P2 states, it does not affect the on/off status of any module.
- Resume: Automatically activates the effect when switching to a module, and restores the module to its preset original switch state when switching away.
- Disable: Automatically activates the effect when switching to a module, and directly turns off the module effect when switching away.

3. PEDAL SWITCH

- Control Knob: KNOB3

Function Description:

This function is the same as the PEDAL SWITCH on the CTRL interface. Please note that this parameter is not a global feature, it is a convenient toggle of the P1/P2 states and is put here for ease of use only.

Global EQ Interface

Core Function

Optimizes the global frequency characteristics of the device's native effect tones, enhancing the balance and expressiveness of the output sound as a whole. It only affects the tones processed by the device's native effects and does not impact the output of external audio sources such as BT, USB, DRUM.

I. Basic Control Methods

Control Object	Operation Method	Function Description
Global EQ Switch	Knob1 Knob / Short Press C Footswitch	Instantly activate or deactivate the Global EQ function, When it is turned on, it becomes a green curve; when it is turned off, it becomes a white curve.
Low Cut	Knob2 Knob	Adjust the low-cut frequency (unit: Hz) to filter low-frequency noise and reduce muddiness
High Cut	Knob3 Knob	Adjust the high-cut frequency (unit: Hz) to filter high-frequency noise and minimize harshness

II. Four-Band (P1-P4) Fine-Tuning Parameters

The Global EQ includes four independently adjustable bands (P1-P4). Each band can be precisely optimized through three parameters: "Center Frequency, Q, and Gain." The specific definitions are as follows:

Parameter Symbol	Name	Unit	Function Description
HZ	Center Frequency	Hertz (Hz)	Sets the core adjustment point of the frequency band (e.g., select 80Hz for low-frequency band, 1kHz for mid-frequency band), clearly defining the target frequency for optimization
Q	Q	-	Controls the bandwidth of the frequency band: A higher Q value results in a narrower adjustment range (precisely targeting the frequency); a lower Q value results in a wider adjustment range (covering surrounding frequencies)
G	Gain	dB (Decibel)	Adjusts the intensity of the frequency band: Positive values enhance (e.g., +3dB thickens the low frequencies); negative values attenuate (e.g., -2dB suppresses harshness in high frequencies)

In short: P1-P4 function as four "Precision Tuning Units." By combining the HZ, Q, and G parameters, they allow targeted optimization of low, mid, and high-frequency details, enabling personalized sound customization.

III. Usage Notes

- 1.Parameter Control:** When adjusting the gain, control the amplitude to avoid sound distortion caused by excessive gain, while protecting the device's speakers and personal hearing.
- 2.Function Limitations:** The global EQ does not affect USB audio or BT audio output, it doesn't affect the sound of the drums; it only applies to the device's effect tones.
- 3.System Load:** If the current preset loads multiple layers of effects (such as multi-segment reverb or delay), enabling the global EQ may increase the system load, causing sound stuttering or overload. It is recommended to simplify the effect chain before use.
- 4.Parameter Backup:** The global EQ parameters can be imported/exported via the device's companion phone or computer software, facilitating the backup of commonly used settings and quick switching between scenarios such as performances and practice.

MIDI Interface Description

Core Function

Used to view and configure the device's MIDI control-related information.

- CC Code List: Display / set Control Change codes (CC codes).
- PC Code List: Display / set Program Change codes (PC codes).

SWITCH PRESET INTERFACE

Functional Positioning

Used to select the operation mode for switching presets on the main interface. Two modes are provided to adapt to different usage scenarios (such as quickly switching tones or accurately preselecting to avoid accidental touches).

Mode Switching Method

Rotate the PARA knob or briefly press the C footswitch to quickly switch between the two preset switching modes.

Mode Details

- **Direct Mode:** The selected preset takes effect and is applied immediately, suitable for scenarios requiring rapid tone switching (e.g., changing tones instantly during a performance to ensure smooth transitions).
- **Preselect Mode:** The selected preset enters a "preselection state" and only takes effect after a secondary confirmation action (e.g., triggering an additional confirmation command). This mode is suitable for scenarios where accidental switching must be avoided (e.g., previewing fine adjustments during practice to prevent sudden tone changes due to operational errors).

OTHER INTERFACE DESCRIPTION

Functional Positioning

Used to configure auxiliary function parameters such as device channel output, USB audio processing, recording volume, and interface theme color, adapting to different usage scenarios and personal preferences.

I. Parameter Details

1. RCH (Right Channel Output Mode): Controls the difference in output content between the left and right channels, with 3 levels:

- **Normal:** Both the left and right channels output the processed signal.
- **Dry:** The left channel outputs wet sound (processed sound with effects), while the right channel outputs dry sound (original unprocessed sound).
- **NoCab:** The left channel outputs sound with CAB (cabinet simulation) effects, while the right channel outputs sound without CAB effects.

2.USB (USB Audio Processing Method): Controls the audio transmission and processing logic of the USB interface, with 4 levels:

- **Normal (Factory Default):** Enables Loopback function, allowing USB recording to capture both accompaniment (USB music, BT music) and instrument audio simultaneously, suitable for mobile recording scenarios;
- **NO:** Disables Loopback function, the USB recording won't pick up the accompaniment;
- **Re-amp:** Activates Re-Amp function, enabling dry signal tracks from the PC to be reprocessed through the device's effect modules into wet signal tracks (when enabled, instrument input via the INPUT jack is disabled. So at this moment, if you play the instrument in your hand, the sound will be blocked.
- **Dry:** Activates the "Record Dry, Monitor Wet" function, optimized for recording with computer DAW software — only the dry guitar signal from the INPUT jack is recorded, without accompaniment, facilitating post-production Re-Amp processing.

* Tips:

Loopback function, a common function on sound cards.

When the function is "Normal" gear, the music in the computer/mobile phone will be recorded at the same time with the instrument sound, it's the same track mixed recording, including USB and BT music.

If the function is "No" gear, when using the TANK PRO as a sound card for recording, only the sound of IN-interface and 10 modules can be recorded, which is more convenient for sub-track recording of instrument sounds during computer recording.

3.USBRec (Recording Volume)

- Adjustment Range: 0–100 (higher values correspond to louder recording levels);
- Factory Default: 50 (equivalent to the 0dB reference level).

4.Themes (Theme Colors) Offers 7 different interface theme colors, which can be switched according to usage scenarios or personal preference.

BT: You can turn BT on or off here, which is the same function as long-pressing HOME button.

INPUT LEVEL: Adjusts the input gain of the INPUT jack. The adjustable range is -15 dB to +15 dB.

Active LED Interface Overview

- 1.H=HUE:** The Hue parameter is controlled via Knob 1 and is used to adjust the color spectrum.
- 2.S=Saturation:** The Saturation parameter is controlled via Knob 2 and determines the intensity or vividness of the color.
- 3.V=Value:** The Value parameter is controlled via Knob 3 and governs the overall luminance (brightness/darkness) of the output.

Using the Re-Amp Function in Your DAW to Record or Adjust Tones:

1. Create a new multi-track session in your DAW, import the backing track into Track 1. It is recommended to adjust the MUSIC parameter from 0 to 100 during playback, carefully listening to find a suitable backing track volume level. Crucially, once set, leave the MUSIC at 100. Adjust the final backing track level using the DAW's fader for Track 1 to your preferred listening volume.
2. Navigate to Global > Other > USB. Select the "Dry" position. This configures the USB virtual output channels (Left/Right) to send the unprocessed guitar signal (dry signal).
3. Import the guitar dry signal into Track 2 of your DAW, or you can record the guitar dry signal onto Track 2 while playing along with the backing track on Track 1. This Track 2 guitar dry signal will be the source audio for the subsequent Re-Amp process.

Note: Before recording the guitar, set the "USB Rec" parameter to 50 to ensure the recorded dry signal level is optimal and unaltered.

4. Navigate to Global > Other > USB, select the "Re-Amp" position, mute Track 1 and play only the guitar dry signal from Track 2, arm Track 3 for recording. This captures the processed guitar signal (wet signal) you heard while originally recording the dry track.

Critical: During this Re-Amp process, ensure all preset parameters and Master parameter remain unchanged from the settings used just now.

5. After the first Re-Amp is complete, navigate to Global > Other > USB and select the "Normal" position, play Tracks 1 and 3 (processed guitar) together to hear the final mix.
6. If you are unsatisfied with the processed guitar sound on Track 3, navigate to Global > Other > USB and ensure "Normal" is selected. Then play only Track 1 while playing your guitar, at the same time you browse presets on the TANK PRO or adjust Preset parameters to find a tone you prefer.
7. To perform a second Re-Amp: Select the "Re-Amp" position, mute Track 1 and play only the original guitar dry signal from Track 2, arm Track 4 for recording, then you will obtain the wet sound of the guitar after the second Re-Amp. Finally, switch back to the "Normal" position, review the new mix (Tracks 1 & 4), then you will get the work that suits you best.

VII. Module Effects List

Module Name	Number of Effects	Specific Effects
WAH	6 + 8 Types	Requires Expression Pedal Control (EXP): X-Wah, Funk-Wah, Slide-Wah, Cry-Wah Automatic Control (auto): Wah-Wah, Sense-Wah Compression Type: Compress, Compress Pro, F Compress Drive Type: Boost, A Boost, B Boost, E Boost, Boost_ED
FX1&FX2	20 types each (40 types total)	Gate Type: Noise Gate++, AI Gate MS, Soft Gate, Hard Gate, Pro Gate Compression Type: Compress, Compress Pro, F Compress Drive Type: Boost, A Boost, B Boost, E Boost, Boost_ED Other Types: Lofi, Pitch, Pitch shifter, Octave, Ring External Expression Pedal Control Type: Exp_Vol, Exp_Pan
DS	30 Types	Supports loading AM4 DATA files, containing 30 types of pedal simulation effects
AMP	90 Types	Supports loading AM4 DATA files, containing 90 types of amplifier simulation effects
CAB	90 Types	Supports loading WAV files (1024/2048 sample points), containing 90 types of cabinet simulation effects
EQ	3 + 5 Types	Guitar EQ 6 (6-band guitar equalizer) Bass EQ 7 (7-band bass equalizer) Normal EQ 10 (10-band general equalizer) Other Types: Lofi, Pitch, Pitch shifter, Octave, Ring
MOD	20 Types	Chorus Type: Chorus, Tri Chorus, Chorus Stereo Flanger Type: Flanger, Tri Flanger, Flanger Stereo Tremolo Type: Tremolo, Tri Tremolo, Opto Tremolo, Tremolo Stereo Phaser Type: Phaser, Phaser Stereo Vibrato Type: Vibrato, Tri Vibrato, Opto Vibrato, Vibrato Stereo Univibe Type: Univibe, Tri Univibe, Vibe Stereo Autofilter Type: Autofilter
DLY	27 Types	Mono Delay: Clean, Modern, Echo, Analog, Duck, Dtype, Tremolo, Filter, Dual, Lofi, Pattern, Ice, Reverse Stereo Delay: PingPong Stereo, Clean Stereo, Modern Stereo, Echo Stereo, Analog Stereo, Duck Stereo, Dtype Stereo, Tremolo Stereo, Filter Stereo, Dual Stereo, Lofi Stereo, Pattern Stereo, Ice Stereo, Reverse Stereo
REV	18 Types	Mono Reverb: Room, Hall, Plate, Spring, Shimmer, Bloom, Cloud, Lofi, Swell Stereo Reverb: Room Stereo, Hall Stereo, Plate Stereo, Spring Stereo, Shimmer Stereo, Bloom Stereo, Cloud Stereo, Lofi Stereo, Swell Stereo

VIII. Technical Specifications

Sample Rate: 44.1KHz/24bit

Dynamic Range: 117dB

Number of Effects: 300+

Effect Modules: Up to 10 simultaneously usable

Number of Presets: 120 (60 factory presets)

Maximum LOOP Recording Time: 150.8s (mono and stereo), automatically saved after power off

Built-in Drum Machine: 100 rhythm patterns

Input Interfaces:

One 6.35mm (1/4") TS instrument input

One 6.35mm (1/4") TRS expression pedal input

One 3.5mm (1/8") TRS MIDI input

Output Interfaces:

One pair of 6.35mm (1/4") TS stereo unbalanced outputs

One pair of XLR stereo balanced outputs with ground lift switch

One 3.5mm (1/8") TRS stereo headphone output

Display: 2.4-inch LCD screen

USB Interface: USB 2.0 TYPE-C, supports USB AUDIO, USB MIDI, LOOPBACK switch, RE-AMP function, built-in battery charging input, data transfer

Dimensions: 260mm (L) × 100mm (W) × 52mm (H)

Weight: Approx. 815 g

Working Current: Approx. 400mA

Battery Rated Capacity: 3000mAh 11.1Wh

Battery Model: DTP105050

Battery Nominal Voltage: 3.7V

Rated Charging Voltage and Current: 5V $\pm$ 1A

Battery Life: Approx. 7 hours

BT Accompaniment: Supported

Community Tone Sharing Platform: Supported

AMP and DS Modules: Support importing amX Data files

CAB Module: Supports importing 44.1KHz/24bit mono WAV files, 1024/2048 sample points

Tone Editing Software Support: WINDOWS, MAC, IOS, ANDROID

Included Accessories: USB cable ×1, user manual ×1

IX. Other

● Mobile APP Connection Procedure: Before operation, ensure both the mobile phone and TANK PRO have their BT functions enabled. Then open the M-EFCS app, tap the "≡" button in the upper left corner, select "Current Device" and choose TANK PRO, then connect and start using.



scan the QR code

● Important Reminder: After each firmware update, the previous BT pairing between the mobile phone and the device may fail. To resolve this, please go to your phone's BT setting interface, first forget the device name of the unit, then reconnect to successfully pair again.

● We continuously strive to improve our products and reserve the right to modify certain features and specifications (including but not limited to appearance, packaging design, operation manual, accessories, dimensions, technical parameters, display, etc.) without prior notice. Please confirm product features and specifications with your local dealer before purchase. Product images and colors may differ from the actual product due to photographic lighting variations and screen settings. Please refer to the physical product.

● Scan the QR code to download the effect modules description:



X. FCC Warning Statement

★ Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment. This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

FCC Radiation Exposure Statement

The device has been evaluated to meet general RF exposure requirement. The device can be used in portable exposure condition without restriction.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- (1) This device may not cause harmful interference.
- (2) This device must accept any interference received, including interference that may cause undesired operation.

一、前言

TANK PRO 是一款集成丰富效果模块、支持灵活自定义、适配多场景的专业效果器，适用于电吉他、电贝司等乐器的音色塑造与演奏创作。

本手册系统涵盖设备的安全规范、面板操作、模式设置、效果参数及自定义流程，旨在帮助您快速掌握设备核心功能，解锁丰富的音色创作可能。

为保障使用安全、充分发挥设备性能，建议您在操作前仔细阅读本手册；若需深入了解特定功能，可查阅对应章节的详细说明。

二、使用须知

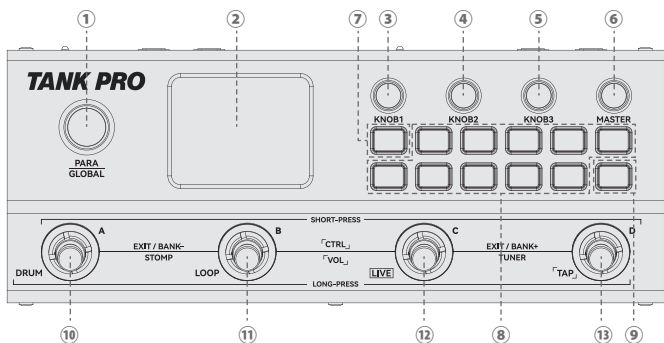
在使用本设备前，请务必仔细阅读以下须知内容：

1. 安全警示：严禁擅自拆解、改装本设备，否则可能造成设备损坏，或引发人身安全风险。
2. 电源管理：设备配备内置电池，支持充电与使用同步进行；当电量不足时，请及时充电，避免设备工作状态不稳定，建议使用5V2A的电源适配器进行充电。
3. 连接操作：在连接或断开本设备与其他设备的线路前，须先关闭相关关联设备。
4. 抗干扰要求：收音机、电视机等设备产生的电磁波可能对本设备造成干扰，建议将本设备与此类设备保持足够距离。
5. 放置环境要求：为避免设备出现变形、变色等损坏情况，请勿将本设备置于以下环境中：
 - a. 各类热源附近；
 - b. 强磁场区域；
 - c. 阳光直接照射的位置；
 - d. 高温潮湿的环境；
 - e. 多粉尘、不洁的场所；
 - f. 存在强烈震荡、摇晃的场景。

三、相关词语解释

- **GLOBAL SAVE**：设备全局设置（通用配置）的保存功能，用于存储设备整体层面的基础运行参数。
- **PATCH SAVE**：单个预设（包含效果程序、音色参数等）的保存功能，用于存储特定音色方案对应的参数配置。
- **SYNC**：设备内部各功能模块的参数同步功能，用于保障各模块间的参数保持匹配一致。
- **VOL 模式**：预设音量调节模式，用于独立调节对应预设的输出音量。
- **CTRL 模式**：踩钉功能自定义模式，支持根据实际使用需求，个性化配置设备踩钉的功能。
- **TAP 模式**：预设 BPM（节拍速度）调节模式，用于设定或调整对应预设的节拍速度参数。
- **LIVE 模式**：此模式适用于现场演出，屏幕上会显示出 4 个踩钉的预设名称和预设号，便于在演出时随时查看当前的 4 个预设名称，而在切换 Bank 的时候也可以查看目标的 4 个预设名称。
- **短踩Short Press**：踩下动作和抬起动作在1.5s 内完成，抬起的一瞬间即生效。
- **长踩Long Press**：踩住踩钉，超过1.5s后才抬起，在 1.5s 的这一瞬间生效。

四、面板介绍



主面板部件说明

1. PARA/GLOBAL 旋钮（带按压功能旋转编码器）

- 短按：进入 / 退出项目、执行确认操作；
- 旋转：选择项目、调节参数、移动光标；
- 长按：进入 GLOBAL 全局设置界面，退出该界面后全局设置将自动保存。

2. 显示屏：2.4 英寸 LCD 屏，用于显示设备当前工作状态。

3. KNOB1 自定义旋钮：出厂默认用于调节 AMP 模块的 Gain 参数。

4. KNOB2 自定义旋钮：出厂默认用于调节 AMP 模块的 Level 参数。

5. KNOB3 自定义旋钮：出厂默认用于调节 AMP 模块的 Middle 参数。

6. MASTER 旋钮：输出音量调节旋钮(作用于IN接口和DRUM和LOOP，不作用于USB和BT，BT&USB VOL 请到HOME界面调节MUSIC参数即可)。

* 请注意：

当您用手机BT或者电脑USB连接本机进行音乐播放时，需要提前把电脑或者手机的音量调小，或者把 MUSIC参数调小，避免音量过大。

7. HOME 带灯按键/BT开关键(带BT状态灯)

- 长按：控制 BT 开关（未连接设备时闪烁，已连接时常亮，灯熄灭时代表 BT 已被关闭）；
- 主界面短按：进入 STOMP 界面，再次短按返回主界面。

8. 功能模块带灯按键（对应 WAH/FX1/FX2/DS/AMP/CAB/EQ/MOD/DLY/REV 模块）

- 短按：进入对应模块界面，再次短按开启模块；
- 长按：控制模块开关；
- 模块开启后：按键常亮。

9. SAVE 带灯按键

- 短按：进入预设保存界面，界面内短按完成 PATCH SAVE；
- 预设参数改动后：按键自动闪烁，SAVE按键灯会通过闪烁来提示当前的预设参数是否已经被改动，如果灯闪烁即代表参数已经被改动，在切换预设之前请及时保存，以免当前的数据被丢失
- 长按：完成 GLOBAL SAVE。

10. A 带灯踩钉：长踩进入 DRUM 界面，再次长踩返回主界面。

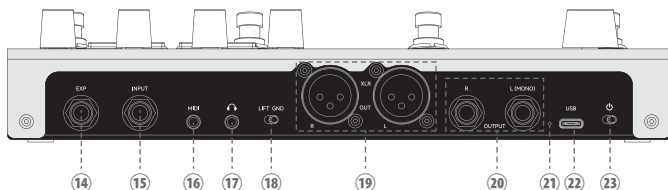
11. B 带灯踩钉：长踩进入 LOOP 界面，再次长踩返回主界面。

12. C 带灯踩钉：长踩进入 LIVE 模式，再次长踩返回主界面。

13. D 带灯踩钉：长踩进入TAP模式，再次长踩返回主界面。

* 组合踩钉操作

1. 长踩 A+B: 进入 STOMP 界面, 再次长踩返回主界面。
2. 长踩 C+D: 进入 TUNER 界面, 再次长踩返回主界面。
3. 长踩 B+C: 进入预设 VOL 模式。
 - 该模式下: 短踩 / 长踩 B 降低音量、短踩 / 长踩 C 增大音量。
 - 退出方式: 长踩 B+C。
4. 短踩 B+C: 进入 CTRL 踩钉自定义界面, 再次短踩返回上一界面。
5. 短踩 A+B 或 C+D (在非主界面和LIVE界面的情况下): 返回主界面。
6. 短踩 A+B 或 C+D (在主界面和LIVE界面的情况下): 用于预设切组。



接口与附属部件说明

14. EXP 接口: 规格为 6.35mm (1/4") TRS 接口, 用于接入外接表情踏板(请使用1/4"立体声音频线)。踏板可设置P1/P2状态, 使用两个状态, 即可用一个踏板调节两个不同状态的参数。可自定义P1/P2的参数, 还可在CTRL界面用踩钉来切换踏板的P1/P2状态。

15. INPUT 接口: 规格为 6.35mm (1/4") 单声道输入接口, 用于连接电吉他、电贝司等乐器。

16. MIDI 接口: 规格为 3.5mm (1/8") TRS 接口, 用于接入 MIDI 控制器。

17. 耳机接口: 规格为 3.5mm (1/8") 立体声输出接口, 用于连接耳机。

18. LIFT/GND 拨档

为 XLR OUT 浮地 / 接地拨档: 拨至“LIFT”档位对应浮地状态, 拨至“GND”档位对应接地状态。

19. XLR OUT 接口: 为 XLR 卡侬平衡输出接口, 单声道输出可连接左声道 (L), 用于对接调音台、声卡等设备。

20. OUTPUT 接口: 规格为 6.35mm (1/4") TS 非平衡输出接口, 单声道输出可连接左声道 (L/MONO), 用于对接乐器音箱等设备。

21. 充电指示灯: 开机正常运行时处于熄灭状态; 设备充电过程中亮起红色; 电量充满后切换为蓝色。

22. USB 接口 (USB TYPE-C)

● 基础功能: 为设备电池充电; 连接上位机 (可实现参数编辑、音色备份、DS/AMP/CAB 音色导入、固件更新、恢复出厂设置);

● 拓展功能: 支持 USB MIDI、USB AUDIO、电脑声卡功能, 同时支持电脑数据传输、OTG 内录;

● 温馨提示: 使用本机附带 USB 线 + 另行选购的 OTG 转换器, 可连接手机第三方 APP, 实现音画同步实时内录。

23. 电源开关: 用于控制设备的开启与关闭。

1. KNOB 旋钮自定义

映射步骤

- ①进入效果模块界面后，通过 PARA 旋钮选中目标控制参数，随后短按 PARA 旋钮；
- ②参数选定框呈现变色状态时，转动 KNOB 旋钮，待弹窗提示出现，即完成映射；
- ③再次短按 PARA 旋钮退出选定状态，KNOB 旋钮即可实现对该参数的自定义控制。

2. CTRL 踩钉自定义

界面说明

CTRL 界面上方 4 个功能格，对应 A、B、C、D 踩钉的短踩功能；界面下方 4 个功能格，对应 4 个踩钉的长踩功能。（为了兼顾便捷性和简洁性，目前只可自定义单个踩钉的功能，因为组合踩钉的功能已经被固定设计好了）。

映射步骤

- ①进入 CTRL 界面，通过 PARA 旋钮选中目标踩钉；
- ②短按 PARA 旋钮进入功能选择界面，选中需映射的功能；
- ③短按 PARA 旋钮确认，设备自动返回 CTRL 界面，即完成映射。

踩钉映射功能表

短踩	长踩	说明
MODE CTRL	MODE CTRL	多模块开关
PEDAL SWITCH	PEDAL SWITCH	EXP 外接表情踏板状态切换（P1/P2）
DLY FREEZE	DLY FREEZE	延迟效果冻结（须在 CTRL 界面才能维持冻结效果，更换预设或者退出 CTRL 界面后冻结效果立即失效。）
KILL SWITCH	KILL SWITCH	断音（多次短踩：瞬间断音；长踩：无声，松开恢复），映射占上下两格
PRESET+1	PRESET+1	预设 + 1
PRESET-1	PRESET-1	预设 - 1
PRESET+4	PRESET+4	预设 + 4（预设跳组）
PRESET-4	PRESET-4	预设 - 4（预设跳组）
VOL+5	VOL+5	预设音量 + 5
VOL-5	VOL-5	预设音量 - 5
BPM+1	BPM+1	预设 BPM+1
BPM-1	BPM-1	预设 BPM-1
MOD TAP	-	调节 MOD 模块 SPEED 参数，连续短踩打点定速
DLY TAP	-	调节 DLY 模块 TIME 参数，连续短踩打点定速
DRUM TAP	-	调节DRUM BPM，连续短踩打点定速
BPM TAP	-	调节预设 BPM，连续短踩打点定速
LOOP	LOOP	启动LOOP录音或者播放
DRUM	DRUM	DRUM 开关
DRUM+1	DRUM+1	DRUM节奏 +1
DRUM-1	DRUM-1	DRUM节奏 -1
BYPASS	BYPASS	旁通模式
MUTE	MUTE	静音模式
PATCH SAVE	PATCH SAVE	保存预设设置
LOOP STOP	LOOP STOP	停止LOOP
NONE	NONE	清空该踩钉映射功能

3. EXP 外接表情踏板自定义

前提条件

若需用外接踏板控制模块参数，插入踏板后，需先进入 GLOBAL 下的 PEDAL 界面完成踏板校准，校准完毕后，即可进行以下的“映射步骤”。

映射步骤

- ①进入效果模块界面，通过 PARA 旋钮选中待控制的参数，短按 PARA 旋钮；
- ②参数选定框变色后，踩动外接踏板，弹窗提示即完成映射；
- ③再次短按 PARA 旋钮退出选定状态，此时 EXP 外接表情踏板可自定义控制该参数。

P1/P2 状态切换与保存

- 若需在设置 P1 状态后继续设定 P2，需进入 CTRL 界面或者进入"GLOBAL-PEDAL-SWITCH" 即可切换踏板状态，再重复上述映射步骤；
- 踏板的 P1、P2 状态会随预设保存：当前预设中踏板处于 P1 状态时，保存该预设后，下次调用此预设，踏板默认仍为 P1 状态。

4. 提示

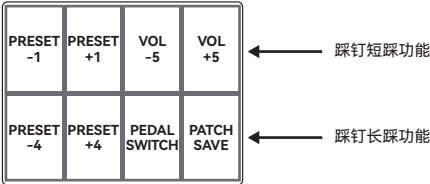
前述 KNOB 旋钮、CTRL 踩钉、外接表情踏板的自定义设置，均与对应预设绑定并随预设保存,执行 PATCH SAVE 操作时，此类自定义设置将与该预设的其他参数同步存储；切换预设时，对应预设的自定义是设置也会随之同步生效。

六、模式 & 界面介绍及操作逻辑

* 温馨提示：

关于在主界面使用踩钉进入和退出各种操作界面的设计理念，我们设计为：以某一或者两个踩钉的长踩或者短踩进入某个界面后，再以同样的方式，即可退出此界面。

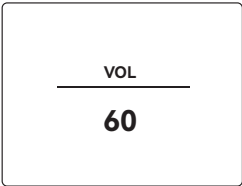
CTRL 模式界面



- 1. 进入方式：任意界面短踩 B&C 组合踩钉，即可进入 CTRL 模式。
- 2. 核心特性：仅 CTRL 模式下，踩钉自定义功能生效（出厂默认设置如上图所示）。
- 3. 踩钉功能分布

踩钉 / 操作	A	B	C	D	A&B	B&C	C&D
短踩	自定义	自定义	自定义	自定义	退出当前界面	退出当前界面	退出当前界面
长踩	自定义	自定义	自定义	自定义	-	-	-

VOL 模式



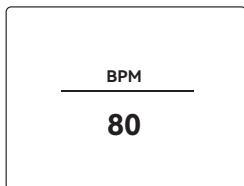
- 1. 进入方式：在非CTRL模式下，任意界面长踩 B&C 组合踩钉，即可进入 VOL 模式。
- 2. 核心特性：VOL 模式是预设专属的音量调节模式，支持通过指定踩钉实现“单次固定幅度音量调节”与“连续音量调节”，是演奏场景下快速、精准控制预设音量的专用功能模式。
- 3. 踩钉功能分布：

踩钉 / 操作	A	B	C	D	A&B	B&C	C&D
短踩	-	单次降低音量	单次降低音量	-	退出当前界面	进入CTRL模式	退出当前界面
长踩	-	连续降低音量	连续降低音量	进入TAP模式	-	退出当前界面	-

4. 模式提示

- 进入 VOL 模式后，仅 B、C 踩钉指示灯亮起，直观提示当前处于预设音量调节状态。

TAP 模式



- 1. 进入方式：**在非CTRL模式下，任意界面长踩 D 踩钉，即可进入 TAP 模式。
- 2. 核心特性：**TAP 模式是与预设绑定的专属 BPM 调节功能模式，支持“打点定速”操作方式，可快速匹配演奏者的实时节奏；若对应效果模块开启 SYNC 功能，模块将与 TAP 模式设定的 BPM 同步运行，是现场演出、创作过程中同步曲目节拍的核心功能。

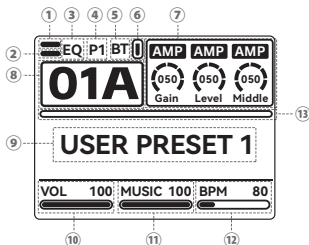
3. 踩钉功能分布：

踩钉 / 操作	A	B	C	D	A&B	B&C	C&D
短踩	-	单次降低BPM	单次降低BPM	连续短踩打点定速(设定BPM)	退出当前界面	进入CTRL模式	退出当前界面
长踩	-	连续降低BPM	连续降低BPM	退出 TAP 模式	-	进入VOL模式	-

4. 模式提示：

- 进入 TAP 模式后，D 踩钉指示灯随当前 BPM 速度同步闪烁，直观反馈节奏速度。

主界面



一、显示内容

- 1.INPUT 接口信号显示：**显示 INPUT 接口传入的实时音频信号状态。
- 2.混合输出信号显示：**显示设备混合输出的实时音频信号状态。
- 3.全局 EQ 状态显示：**开启后常亮，关闭后熄灭。
- 4.EXP 踏板状态显示：**显示外接表情踏板工作模式（P1/P2），状态变更需通过 PATCH SAVE 保存。
- 5.BT 状态显示：**BT状态独立保存，状态改动后会自动保存。
 - 连接成功：常亮；
 - 开启未连接：闪烁；
 - 未开启：熄灭。
 (BT状态跟HOME键闪烁状态一致)
- 6.电量显示。**
- 7.自定义旋钮控制参数显示（KNOB1/KNOB2/KNOB3）：**
 - 出厂默认：分别对应 AMP 模块 Gain、Level、Middle 参数；
 - 参数映射变更需通过 PATCH SAVE 保存。
- 8.预设编号显示：**
 - 分组逻辑：BANK 分组（01 为第一组），每组 A、B、C、D 对应 4 个音色；
 - 存储容量：30 组共 120 个预设储存位；
 - 出厂分配：01-15组（出厂自带音色）、16-30 组（无效果音色）。

*** 提示:**

当您切换预设后，预设位置并不会自动保存，如果需要存储位置，在关机前，长按SAVE键，然后屏幕会弹出“Global Save”，即可在下次开机时得到您关机前的预设位置。

9.预设名称显示: 可在 SAVE 界面修改，修改后需通过 PATCH SAVE 保存。

10.VOL (预设音量) 显示: 显示当前预设音量参数，变更需通过 PATCH SAVE 保存。

11.MUSIC 音量显示: 显示 BT、USB 输入音源音量参数，变更需通过 GLOBAL SAVE 保存。

12.BPM (预设节拍) 显示: 显示当前预设 BPM 参数，变更需通过 PATCH SAVE 保存。

13.Looper 展示条: 如果 Looper 里面存在音频的情况下，此展示条会显示在主界面。

*** 预设BPM和效果SYNC开关的关联**

核心作用: 控制效果参数Time/Speed、鼓机BPM与预设 BPM 的关联，实现效果速度与音乐节拍的同步或独立控制。

*** SYNC 关闭状态**

- 参数单位: Time (延迟时间、颤音周期等)、Speed (滤波扫描速度、LFO 速率等) 以绝对时间单位 (如 ms) 设定;
- 控制方式: 手动调节参数，与预设 BPM 无关联;
- 效果特点: 效果速度固定，不受预设 BPM 变化影响。

*** SYNC 开启状态**

- 参数单位: Time、Speed 切换为相对节拍单位 (如 1/4 拍、1/8 拍等，含附点 / 三连音细分节拍) ;
- 控制方式: 参数实际速度随预设 BPM 自动计算;
- 效果特点: 效果速度与音乐节拍同步，预设 BPM 变化时自动调整。

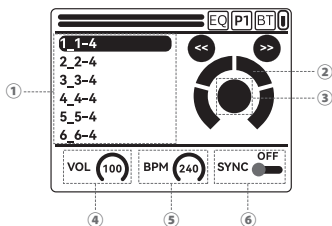
二、踩钉功能

主界面下，单个及组合踩钉的长 / 短踩功能固定，具体分布如下：

踩钉 / 操作	A	B	C	D	A&B	B&C	C&D
短踩	BANK (组内 A 预设)	BANK (组内 B 预设)	BANK (组内 C 预设)	BANK (组内 D 预设)	BANK- (组-1)	进入CTRL模式界面	BANK+ (组+ 1)
长踩	进入DRUM界面	进入LOOP界面	进入LIVE模式界面	进入TAP模式	进入STOMP界面	进入VOL模式	进入TUNER界面

DRUM 界面

进入方式: 在非CTRL模式下，任意界面长踩 A 踩钉，即可进入 DRUM 模式。



一、界面显示内容

1.鼓机节奏型列表: 展示可选择的鼓机节奏类型 (100个)。

2.拍数显示: 显示当前鼓机运行的拍数。

3.第一拍提示: 第一拍时变绿，辅助节奏对齐。

4.DRUM 音量: 调节鼓机输出音量。

5.DRUM BPM。

6.同步开关: 控制鼓机 BPM 与预设 BPM 是否同步。

*** 提示:**

DRUM界面内的设置可通过GLOBAL SAVE 保存。

二、踩钉功能

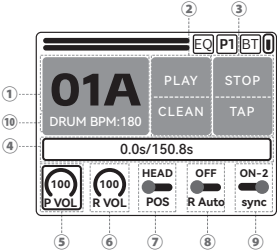
踩钉 / 操作	A	B	C	D	A&B	B&C	C&D
短踩	DRUM节奏-1	DRUM节奏+1	DRUM开关	调节DRUM BPM，连续短踩打点定速	退出 DRUM 界面	进入CTRL模式界面	退出 DRUM 界面
长踩	退出 DRUM 界面	进入LOOP界面	进入LIVE模式界面	进入TAP模式	进入STOMP界面	进入VOL模式	进入TUNER界面

三、踩钉灯指示规则

- 鼓机开启时，C 踩钉灯常亮；鼓机关闭时，C 踩钉灯熄灭。
- 处于鼓机 DRUM 界面时，D 踩钉随鼓机 BPM 速度闪烁，直观反馈节拍节奏。

LOOP 界面

进入方式：在非CTRL模式下，任意界面长踩 B 踩钉，即可进入 LOOP 模式。



一、显示内容

- 1.预设编号显示：展示当前预设编号。
- 2.C 踩钉功能显示：上方为短按功能，下方为长按功能。
- 3.D 踩钉功能显示：上方为短按功能，下方为长按功能。

①REC: Record 启动录音，当 LOOPER 里面没有音乐时候，会显示 REC。

②PLAY: 当音乐停止时，踩这个踩钉可播放音乐；在 Double 叠加录音时，踩这个踩钉可以停止录音并且播放音乐。

③CLEAN: 当音乐停止时，或者正在录制第一层音频时，长踩这个踩钉可以 delete删除音乐，清空 LOOPER 的音乐。

④STOP: 单踩第四个踩钉可以停止录音并且保存，或者停止播放。

⑤TAP: 长踩第四个踩钉可以进入 TAP 界面。

⑥BACK: Back就是返回的意思，可无限次数重做或撤销最后一次叠加录制的音轨(Undo/Redo)。

⑦DUB: Double叠加录音，这个功能可实现无限叠加音轨录音。

- 4.LOOP 时长显示：显示循环总时长（最长支持 150.8 秒）。
- 5.LOOP 播放音量：调节循环播放的输出音量。
- 6.LOOP 录音音量：调节循环录音的输入音量。
- 7.LOOP 音链位置档位：选择 LOOP 在效果链中的位置（首端 / 末端）。
- 8.LOOP 自动录音灵敏度档位：分 OFF、HIGH、MID、LOW 四档；功能开启后，您弹琴的一瞬间，LOOP 会上启动录音，从而实现AUTO REC自动录音，无需动脚。HIGH 代表灵敏度很高，轻轻地弹琴即可启动录音。以此类推，LOW 代表灵敏度较低，需要用力弹琴才可以启动录音。

*** 提示：**
必须要在 LOOP 界面才能使用 R Auto 功能。

- 9.DRUM SYNC鼓机同步：开启后，鼓会跟随LOOP同步启动，鼓和LOOP也会智能自动对齐。ON1和ON2的功能区别是，ON2的DRUM会随LOOP暂停，ON1的DRUM不会随LOOP暂停。
如果您需要使用这个功能，请在录制第一层音频之前把此功能打开，即可实现鼓和音频的完美对齐。如果您先录制音频，再去打开这个功能，那可能会导致对齐失败。

*** 提示：**
LOOP界面内的设置可通过GLOBAL SAVE 保存。

- 10.当 DRUM SYNC 功能被开启后，此处会显示出鼓机的速度。

二、界面踩钉功能

踩钉 / 操作	A	B	C	D	A&B	B&C	C&D
短踩	预设 - 1	预设 + 1	功能实时变动， 请看下表	停止LOOP	退出 LOOP 界面	进入CTRL模 式界面	退出 LOOP 界面
长踩	进入DRUM 界面	退出LOOP 界面	功能实时变动， 请看下表	进入TAP 模式	进入STOMP 界面	进入VOL模 式	进入TUNER 界面

*** 温馨提示：**
在Loop界面时，建议不使用右边第3个踩钉与其他踩钉的组合功能，比如上图的B&C和C&D。因为Loop界面下的第三个踩钉功能，是踩下即马上生效，所以组合踩钉的操作会引起踩钉功能冲突。

C 踩钉操作逻辑	
操作场景	功能效果
LOOP有内容，暂停状态下，单踩踩钉	播放LOOP音频
LOOP无内容时，单踩踩钉	启动 LOOP 录音
录音完成单踩踩钉	启动 LOOP 播放
播放状态下单踩踩钉	叠加音轨
录制 / 播放时，长踩踩钉	删除 / 撤销 / 恢复音轨（单音轨时为“删除”；叠加音轨时为“撤销 / 恢复最后一次录音”）
录制 / 播放时，双踩踩钉	暂停 LOOP（录音 / 播放暂时停止）
暂停后长踩踩钉	删除当前档位下所有录音数据

三、踩钉指示灯规则

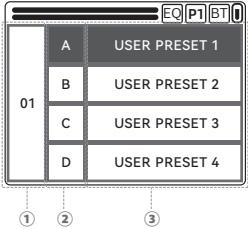
- LOOP 录音轨无内容时，C 踩钉灯不亮；
- 录音轨有内容时，C 踩钉灯亮起； LOOP 基本操作过程中，灯色与 C 踩钉功能显示区颜色保持同步。

A踩钉

- 当鼓机开启时，A 踩钉灯会与鼓机当前设定的 BPM 节奏同步闪烁；
- 若鼓机关闭，A 踩钉灯保持长亮状态。

LIVE 模式界面

进入方式：在非CTRL、LOOP模式下，任意界面长踩 C 踩钉，即可进入 LIVE 模式。



一、界面显示

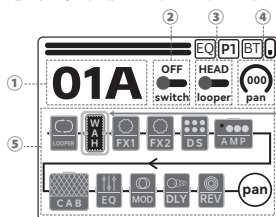
- 1.预设组号编号：显示当前预设所属分组的编号。
- 2.预设对应踩钉：标识与当前预设关联的踩钉。
- 3.预设名称：展示当前预设的自定义名称。

二、踩钉功能

踩钉 / 操作	A	B	C	D	A&B	B&C	C&D
短踩	BANK（组内A预设）	BANK（组内B预设）	BANK（组内C预设）	BANK（组内D预设）	BANK-（组-1）	进入CTRL模式界面	BANK+（组+1）
长踩	进入DRUM界面	进入LOOP界面	退出LIVE模式界面	进入TAP模式	进入STOMP界面	进入VOL模式	进入TUNER界面

STOMP 界面

进入方式：在非CTRL、VOL、TAP模式下，任意界面长踩 A+B 踩钉，即可进入 STOMP 模式。



* 便捷操作提示：

对当前预设进行 save 操作后，这个白色框所在的位置会跟随预设进行保存。因此用户可用踩钉进入 stomp 模式然后踩第三个踩钉即可即兴开/关某个模块，而不需要到 CTRL 界面提前设置踩钉功能。

一、界面显示

1. 预设编号：显示当前预设。
2. 模块开关档位：由 KNOB1 旋钮控制，调节模块的开关状态。
3. LOOP 效果链位置档位：由 KNOB2 旋钮控制，选择 LOOP 在效果链中的位置（首端 / 末端）。
4. 左右声道信号输出比例：由 KNOB3 旋钮控制，调整左右声道输出占比。
5. 效果链模块顺序：通过 PARA 旋钮选定模块，调节其在效果链中的排列顺序。

* 提示：

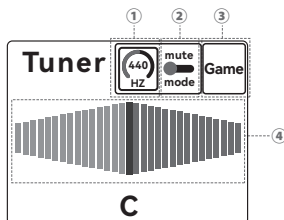
1. STOMP 界面内的 pan 参数、白框光标位置、效果模块顺序可通过 PATCH SAVE 保存。
2. FX 模块中的 Exp_Pan 属于立体声效果，与 MOD（调制）、DLY（延迟）、REV（混响）模块中的立体声效果同属一类，核心均基于双声道信号的相位差与时间差构建空间感，而单声道效果仅针对单通道信号进行独立处理，两者信号处理逻辑存在本质区别；因此所有立体声效果（含 Exp_Pan 及 MOD/DLY/REV 的立体声效果）须置于效果链末端（信号输出前的最终处理环节），若提前接入，后续单声道效果会破坏双声道空间相位关系，导致声音浑浊、定位偏移或空间感失效等异常。

二、踩钉功能

踩钉/操作	A	B	C	D	A&B	B&C	C&D
短踩	预设 - 1	预设 +1	模块开关	光标循环选择模块（光标位置随预设储存）	退出 STOMP 界面	进入 CTRL 模式界面	退出 STOMP 界面
长踩	进入 DRUM 界面	进入 LOOP 界面	进入 LIVE 模式界面	进入 TAP 模式	退出 STOMP 界面	进入 VOL 模式	进入 TUNER 界面

TUNER 界面 (GAME 模式)

进入方式：在非CTRL、VOL、TAP模式下，任意界面长踩 C+D 踩钉，即可进入 TUNER 模式。



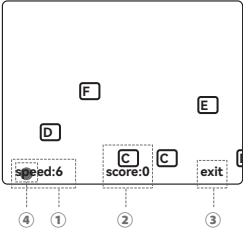
一、界面显示

1. 校音基准频率设定：由 KNOB1 旋钮调节校音基准频率（默认 440Hz，对应国际标准音 A₄ 振动频率），为乐器音高校准提供参考。
2. MUTE/BYPASS 档位：由 KNOB2 旋钮控制，可切换“静音（MUTE）”或“旁通（BYPASS）”工作模式。
3. 游戏开关：由 KNOB3 旋钮控制，用于开启或关闭 GAME 功能。
4. 调音表显示：通过绿、红色条直观反馈音高偏差（色条偏左 = 音高过低，偏右 = 音高过高），下方显示具体音名，辅助精准校音。

二、踩钉功能

踩钉/操作	A	B	C	D	A&B	B&C	C&D
短踩	静音模式	旁通模式	退出TUNER界面	退出TUNER界面	退出TUNER界面	进入CTRL模式界面	退出TUNER界面
长踩	进入DRUM界面	进入LOOP界面	进入LIVE模式界面	进入TAP模式	进入STOMP界面	进入VOL模式	退出TUNER界面

三、GAME 模式界面



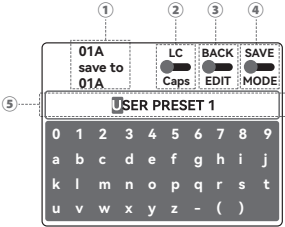
界面显示

- 1.音名滚动速度调节：由 KNOB1 旋钮控制，可调整音名在界面内的滚动快慢。
- 2.得分管理：界面显示 GAME 实时得分，通过 KNOB2 旋钮可清除当前得分。
- 3.游戏退出操作：转动 KNOB3 旋钮，即可退出 GAME 界面。
- 4.识别点反馈：依据 INPUT 接口输入信号的音高，识别点在界面内上下移动，直观反馈音高识别状态。

返回逻辑

在 GAME 界面中，任意短踩 A/B/C/D 单个踩钉，即可直接返回 TUNER 调音主界面。
短踩A&B或C&D踩钉，直接退回主界面。

SAVE 界面



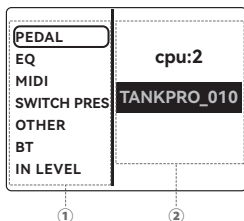
一、界面显示与控制元素

- 1.预设编号及操作显示：直观呈现当前预设编号，同步展示 SAVE（保存）、SWAP（交换）两种操作选项。
- 2.字母大小写切换：由 KNOB1 旋钮控制，可切换输入字母的大小写状态。
- 3.删除 / 空格控制：由 KNOB2 旋钮控制，顺时针转动触发空格输入，逆时针转动触发字符删除。
- 4.操作模式切换：由 KNOB3 旋钮控制，用于切换执行 SAVE（保存）或 SWAP（交换）操作。
- 5.预设名称编辑区：通过 PARA 旋钮选定名称中的单个字符，进入键盘界面选择目标字符完成编辑。

二、保存 / 交换操作步骤

- 1.完成预设名称编辑后，通过 KNOB3 旋钮选择需执行的操作（SAVE 或 SWAP）；
- 2.确认预设的保存 / 交换位置；
- 3.短按 SAVE 按键，确认执行操作；
- 4.操作成功后，界面会显示“Save OK”提示，完成整个流程。

GLOBAL 界面



界面显示元素

1. 设置菜单列表（左侧）：列出 PEDAL、EQ、MIDI、SWITCH PRESET、OTHER、BT、INPUT LEVEL 等设置选项，可选择对应功能进入详细配置页面。

2. 系统信息显示（右侧）：展示 CPU 运行信息，同时呈现设备固件版本标识，方便查看设备状态及版本信息。提示：当负载率超过90（90%）时，可能会导致杂音或者其他问题。

* 温馨提示：

当您退出全局界面的一瞬间，本机将执行一次全局参数的保存，以此可以对参数进行自动保存，这个功能，跟长按SAVE键的功能是一样的。保存成功后会显示“Global save”。

PEDAL（外接表情踏板）界面

核心作用

用于外接表情踏板的校准与功能参数设置，使用外接踏板前必须先进入此界面完成配置。

一、踏板校准步骤

1. 进入 PEDAL 界面后，根据屏幕图案提示完成踏板全行程触发等校准操作；
2. 每完成一个校准步骤，需通过 短按 PARA 旋钮 或 短踩 C 踩钉 确认，直至校准流程完成。

二、参数控制（对应旋钮调节）

1. Invert（踏板正反向档位）

- 控制旋钮：KNOB1
- 功能：切换踏板的控制方向（正向 / 反向），适配不同使用习惯。

2. Toe（踏板 P1/P2 状态切换时的模块开关控制）

- 控制旋钮：KNOB2

档位说明：

- None：踏板切换 P1/P2 状态时，不影响任何模块的开关状态；
- Resume：切换到某模块时自动开启效果，切走时模块恢复预设原始开关状态；
- Disable：切换到某模块时自动开启效果，切走时直接关闭该模块效果。

3. PEDAL SWITCH

- 控制旋钮：KNOB3

功能说明：

此功能跟 CTRL 界面的 PEDAL SWITCH 是同一个功能。请注意，这个参数不属于全局功能，它是预设中的 P1 或 P2 状态的便捷切换键，放在这里只是为了方便操作。

全局 EQ 界面

核心作用

对设备本机效果音色进行全局频率特性优化，整体提升输出声音的平衡度与表现力，且仅作用于本机效果器处理后的音色，不影响BT，USB，DRUM等声音的输出。

一、基础控制方式

控制对象	操作方式	功能说明
全局 EQ 开关	KNOB1 旋钮 / 短踩 C 踩钉	一键开启或关闭全局 EQ 功能，开启后变成绿色曲线，关闭后变成白色曲线
Low Cut（低切）	KNOB2 旋钮	调节低切频率（单位：Hz），过滤低频杂音，减少浑浊感
High Cut（高切）	KNOB3 旋钮	调节高切频率（单位：Hz），过滤高频杂音，降低刺耳感

二、四频段（P1-P4）精细调节参数

全局 EQ 包含 P1-P4 四个独立可调频段，每个频段可通过“中心频率、Q、增益”三参数精准优化，具体定义如下：

参数符号	名称	单位	功能说明
HZ	中心频率	Hertz (Hz)	设定频段核心调整点（如低频段选 80Hz、中频段选 1kHz），明确优化的目标频率
Q	Q 值	-	控制频段带宽：Q 值越高，调整范围越窄（精准定位目标频率）；Q 值越低，调整范围越宽（覆盖周边频率）
G	增益	dB（分贝）	调节频段强度：正值增强（如 +3dB 增厚低频），负值衰减（如 -2dB 压制高频刺耳感）

简言之：P1-P4 相当于四个“精准调音单元”，通过组合 HZ、Q、G 参数，可针对性优化低、中、高不同频段细节，实现个性化音色定制。

三、使用注意事项

- 1.参数控制：调节增益时需控制幅度，避免增益过高导致声音失真，同时保护设备扬声器及个人听力；
- 2.功能限制：全局 EQ 不影响 USB 音频及BT音频输出，也不影响DRUM的声音，仅作用于本机效果音色；
- 3.系统负载：若当前预设加载多层效果（如多段混响、延迟），开启全局 EQ 可能加重系统负载，引发声音卡顿或过载，建议简化效果链后使用；
- 4.参数备份：可通过设备配套手机或电脑软件导入 / 导出全局 EQ 参数，方便备份常用设置，或在演出、练习等不同场景快速切换方案。

MIDI 界面说明

核心作用

用于查看、配置设备的 MIDI 控制相关信息。

- CC 码列表：展示 / 设置控制变化码（CC 码）。
- PC 码列表：展示 / 设置程序变更码（PC 码）。

SWITCH PRESET（预设切换模式）界面

功能定位

用于选择主界面切换预设的操作模式，通过两种模式适配不同使用场景（如快速切换音色或精准预选避免误触）。

模式切换方式

通过 转动 PARA 旋钮 或 短踩 C 踩钉，可在两种预设切换模式间快速切换。

模式详情

- Direct（直接模式）选中预设后立即生效并应用，适用于需要快速切换音色的场景（如演出中即时更换音色，确保衔接流畅）。
- Preselect（预选模式）选中预设后仅处于“预选状态”，需通过二次确认操作（如额外触发确认指令）才生效，适用于需避免误触切换的场景（如练习中精细调整前的预览，防止操作失误导致音色突变）。

OTHER（辅助功能）界面说明

功能定位

用于配置设备的声道输出、USB 音频处理、录音音量及界面主题颜色等辅助功能参数，适配不同使用场景与个人偏好。

一、参数详情

1. RCH（右声道输出模式）控制左右声道输出内容的差异，含 3 个档位：

- Normal：左右声道均输出湿声；
- Dry：左声道输出湿声（经效果处理的声音），右声道输出干声（未经效果处理的原始声音）；
- NoCab：左声道输出带 CAB（箱体模拟）效果的声音，右声道输出无 CAB 效果的声音。

2.USB（USB 音频处理方式）控制 USB 接口的音频传输与处理逻辑，含 4 个档位：

- Normal（出厂默认）：开启 Loopback 功能，USB 录音同时录入伴奏声（USB 音乐、BT 音乐）和乐器声，适配手机录音场景；
- NO：关闭 Loopback 功能，USB 录音时不会录入伴奏声；
- Re-amp：开启 Re-Amp 功能，可将 PC 端干信号轨道通过本机效果模块重新处理为湿声轨道（开启后屏蔽 INPUT 接口乐器输入，弹奏无声音输出）；
- Dry：开启“听湿录干”功能，适配电脑 DAW 软件录音，录音仅记录 INPUT 接口吉他干声，不含伴奏声，方便后期 Re-Amp 处理。

* 温馨提示：

Loopback 功能，是常见的声卡回环功能。功能在 Normal 档位时，电脑/手机内的音乐声会与乐器声同时录制，同一轨道混合录制，其中包括 USB 和 BT 的音乐声。如果在 NO 档位，使用 TANK PRO 作为声卡进行录音时候，只能录制到 INPUT 接口和 10 个模块的声音，这样更便于电脑录音时候进行乐器声分轨录音。

3.USBRec（录音音量）

- 调节范围：0-100（数值越大，录音音量越大）；
- 出厂默认：50（对应 0dB 基准音量）。

4.Themes（主题颜色）提供 7 种不同界面主题颜色，可根据使用场景或个人偏好切换。

BT：这里可以开启或者关闭 BT 功能，跟长按 HOME 键是一样的功能。

INPUT LEVEL：调节 INPUT 接口的输入增益，可调范围为 -15dB 到 +15dB。

ACTIVE LED(LED调色)界面说明

- 1. **H=HUE 色相：**由 Knob1 控制，调整颜色种类。
- 2. **S=Saturation 饱和度：**由 Knob2 控制，调整颜色鲜艳程度。
- 3. **V=Value 亮度：**由 Knob3 控制，调整整体明暗程度。

如何在 DAW 中使用本机 re-amp 功能录制或调节音色：

1. 在 DAW 中新建多轨录音文件，然后在第一轨中导入伴奏文件，此时建议把 MUSIC 参数从 0 调节到 100，慢慢试听一下伴奏声的音量大小。最后务必要把 MUSIC 参数固定在 100，再到 DAW 的第一轨中调节伴奏的音量，选择合适自己的音量。

2. 在 GLOB-Other-USB，选择最后一个档位 Dry，即可实现 USB 虚拟输出通道的左右声道输出的吉他为干音。

3. 在 DAW 的第二个音轨中导入吉他干轨，或者在播放第一轨的伴奏的同时录入吉他干轨。第二轨的吉他干声作为后面 Re-Amp 原始音频。

注意：在录制吉他之前，建议把 USB Rec 参数调节为 50，才能在录制后得到最原本的吉他音量。

4. 在 GLOB-Other-USB，选择 Re-amp 档位，再 mute 第一轨伴奏声，只播放第二轨的吉他干声，然后在第三轨开启录音，即可录制到刚才录制吉他干声时候听到的吉他湿声（注意，在这个 Re-Amp 过程中，要保持效果器的所有 Preset 参数和 Master 参数不被改动，都要跟刚才的是一样的）。

5. 第一次 Re-Amp 完成后，在 GLOB-Other-USB，选择 Normal 档位，然后播放第一轨和第三轨，即可听到最终的成品。

6. 如果您对第三轨的吉他湿声不满意，那在 GLOB-Other-USB，选择 Normal 档位，那现在建议您单独播放第一轨，边听伴奏边弹琴，在 TANK PRO 上切换预设，或者调节 Preset 参数得到您更满意的音色。

7. 选择 Re-amp 档位，再 mute 第一轨伴奏声，只播放第二轨的吉他干声，然后在第四轨开启录音，即可进行第二次 Re-Amp 和录音。最后，切换到 Normal 档位，试听完毕后，即可得到您最满意的作品。

七、模块效果列表

模块名称	效果数量	具体效果
WAH	6 + 8 种	需表情踏板控制 (EXP) : X-Wah、Funk-Wah、Slide-Wah、Cry-Wah 自动控制 (auto) : Wah-Wah、Sense-Wah 压缩类: Compress、Compress Pro、F Compress 激励类: Boost、A Boost、B Boost、E Boost、Boost_ED
FX1&FX2	各 20 种 (共 40 种)	门限类: Noise Gate++、AI Gate MS、Soft Gate、Hard Gate、Pro Gate 压缩类: Compress、Compress Pro、F Compress 激励类: Boost、A Boost、B Boost、E Boost、Boost_ED 其他类: Lofi、Pitch、Pitch shifter、Octave、Ring 外接表情踏板控制类: Exp_Vol、Exp_Pan
DS	30 种	支持加载 AM4 DATA 文件, 包含 30 种单块模拟效果
AMP	90 种	支持加载 AM4 DATA 文件, 包含 90 种音箱模拟效果
CAB	90 种	支持加载 WAV 文件 (1024/2048 采样点), 包含 90 种箱体模拟效果
EQ	3 + 5 种	Guitar EQ 6 (吉他 6 段均衡) Bass EQ 7 (贝斯 7 段均衡) Normal EQ 10 (通用 10 段均衡) 其他类: Lofi、Pitch、Pitch shifter、Octave、Ring
MOD	20 种	合唱类: Chorus、Tri Chorus、Chorus Stereo 镶边类: Flanger、Tri Flanger、Flanger Stereo 颤音类: Tremolo、Tri Tremolo、Opto Tremolo、Tremolo Stereo 移相类: Phaser、Phaser Stereo 震音类: Vibrato、Tri Vibrato、Opto Vibrato、Vibrato Stereo 多谐振荡器类: Univibe、Tri Univibe、Vibe Stereo 自动滤波类: Autofilter
DLY	27 种	单声道延迟: Clean、Modern、Echo、Analog、Duck、Dtype、Tremolo、Filter、Dual、Lofi、Pattern、Ice、Reverse 立体声延迟: PingPong Stereo、Clean Stereo、Modern Stereo、Echo Stereo、Analog Stereo、Duck Stereo、Dtype Stereo、Tremolo Stereo、Filter Stereo、Dual Stereo、Lofi Stereo、Pattern Stereo、Ice Stereo、Reverse Stereo
REV	18 种	单声道混响: Room、Hall、Plate、Spring、Shimmer、Bloom、Cloud、Lofi、Swell 立体声混响: Room Stereo、Hall Stereo、Plate Stereo、Spring Stereo、Shimmer Stereo、Bloom Stereo、Cloud Stereo、Lofi Stereo、Swell Stereo

八、技术参数

采样率：44.1KHz/24bit

信噪比：117dB

效果数量：300+

效果模块：最多同时使用10个

预设数量：120个（60个出厂预设）

LOOP最大录制时间：150.8s（单声道和双声道），关机后自动保存

内置鼓机：100种节奏型

输入接口：

一个6.35mm（1/4"）TS乐器输入接口

一个6.35mm（1/4"）TRS表情踏板输入接口

一个3.5mm（1/8"）TRS MIDI输入接口

输出接口：

一对6.35mm（1/4"）TS 立体声非平衡输出接口

一对卡农（XLR）立体声平衡输出口，附带地线切除开关

一个3.5mm（1/8"）TRS立体声耳机输出接口

显示屏：2.4 英寸 LCD 显示屏

USB接口：USB 2.0 TYPE-C，支持USB AUDIO，支持USB MIDI，支持LOOPBACK开关，支持RE-AMP功能，内置电池充电输入，数据传输

尺寸：260mm（长）×100mm（宽）× 52mm（高）

重量：约815 g

工作电流：约400mA

电池额定容量：3000mAh 11.1Wh

电池型号规格：DTP105050

电池标称电压：3.7V

额定充电电压和电流：5V $\overline{\text{---}}$ 1A

续航时长：约7小时

BT伴奏：支持

社区音色互动平台：支持

AMP和DS模块：支持导入amX Data 文件

CAB模块：支持导入44.1KHz/24bit，单声道WAV文件，1024/2048采样点

音色编辑软件支持：WINDOWS，MAC，IOS，Android

出厂附带：USB连接线×1，说明书×1

修订日期：2026-6-12

九、其他

- 手机 APP 连接步骤：在操作前需确认手机和 TANK PRO 的 BT 功能都已经开启，然后打开 M-EFCS，点击左上角“三”按键，点击“当前设备”选择 TANK PRO，即可连接使用。



扫描二维码下载APP

- 温馨提醒：每次固件升级后，会导致原来手机和机器的 BT 功能配对失败。因此，需要您到手机BT界面，先对机器的设备名进行忽略，然后重新连接，即可成功。
- 我们将不断改进我们的产品并保留修改某些功能和规格 (包括但不限于外观、包装设计、操作手册、配件、尺寸、规格参数、显示屏等) 的权利，恕不另行通知。购买前请与当地经销商确认产品功能及规格。产品图片及颜色因拍照光线误差及屏幕设定可能与实物产品效果有所差异。请以实物为准。
- 扫描二维码下载音色说明：



◆ In the interest of product improvement, the specifications and/or the content of products (including but not limited to appearances, packaging design, manual content, accessories, size, parameters and display screen), are subject to change without prior notice. Please check with local supplier for exact offers. Specifications and features (including but not limited to appearances, colors and size) may vary by model owing to environmental factors, and all images are illustrative.

TANK PRO

Effects Description

DS

Name	Real Amp/Pedal
CT-D805F-OD	Seymour Duncan 805 OD
CT-TS9-OD	Ibanez TS9 Tubescreamer
CT-BOGIE-OD	Mesa Boogie (generic)
CT-5150-DS	MXR 5150 OD
CT-BERZEK-OD	MXR Berzerker Overdrive
CT-BKRFAT-BST	MXR Berzerker Overdrive
CT-BOSS-OD	Boss SD-1 TS808 mod
CT-CHGPRE-DS	Fortin Mesuggah preamp
CT-DST3F-DS	MXR Distortion III
CT-HFAT-FZ	Fuzz CLA
CT-IBZPENTA-DS	Ibanez Penta
CT-MFAT-FZ	Fuzz CLA
CT-OD10-OD	Aria OD-10
CT-TS8F3-OD	Fortin/Ibanez TS808 Tubescreamer
BOS-MT-1	BOSS Metal Zone

Name	Real Amp/Pedal
BOS-MT-2	BOSS Metal Zone
IB-TS8	Ibanez TS808
BOS-DS1	BOSS DS-1
BOS-DS2	BOSS DS-2
M-OD	M-WAVE ODRIVE-AMP
M-DS	M-WAVE DIST-MT1
M-TS1	M-WAVE ODRIVE-TS
M-TS2	M-WAVE ODRIVE-TS
M-SUPA-1	M-WAVE SUPRO
M-SUPA-2	M-WAVE SUPRO
Pro-RAT	Pro Co RAT
Pro-RAT-BST	Pro Co RAT with 808
JHS-AT	JHS AT
JHS-AT-LIMIT	JHS AT+ Andy Timmons
M-BLUES-OD	M-WAVE BLUES-OD

AMP

No.	Name	Real Amp/Pedal
1	CT-RJ20-CL	Roland JC-120
2	CT-JIMI1-OD	PRS HDRX100 (Hendrix Woodstock-era)
3	CT-RH30-OD	Cornford Roadhouse 30
4	CT-RAND666-DS	Randall 667 6ch 120W
5	CT-MPRO51-CL	Mad Professor Old School 51RT
6	CT-CL30-21-OD	Peavey Classic 30
7	CT-CL30-26-CL	Peavey Classic 30
8	CT-RYL-OD	Supro Royale 1932R
9	CT-M-SLO-DS	Myasnikov SuperLead (clone SLO)
10	CT-FENDTW2-CL	Fender Twin Reverb II
11	CT-GASMOL-DS	Diezel D-Moll EL34
12	CT-BLUEVD-OD	Fender Blues Deluxe
13	CT-TGREM-OD	Tone King Gremlin
14	CT-BOG2CP-DS	Mesa Boogie Mark IIC+
15	CT-JAKE20-OD	Friedman JEL-20 (Jake E Lee Signature)
16	CT-UBER58-DS	Bogner Uberschall
17	CT-BAD6L-CL	Mesa Boogie Badlander 6L6
18	CT-CHGPRE-DS	Fortin Mesuggah preamp
19	CT-FENDBLUE-CL	Fender Blues Deluxe Reissue
20	CT-800-OD	Marshall JCM 800 2203
21	CT-STV5-5-BS	Ampeg SVT-5 PRO
22	CT-STV5-1-BS	Ampeg SVT-5 PRO
23	CT-BILLY-5-BS	Ampeg SVT-BSP Billy Sheehan Signature
24	CT-BILLY-1-BS	Ampeg SVT-BSP Billy Sheehan Signature
25	CT-RB8-BS	Gallien-Krueger 800RB
26	CT-800-OD2	Marshall JCM 800 2203

No.	Name	Real Amp/Pedal
27	CT-8100-DS	Marshall 8100 Valvestate
28	CT-BAD6L-DS	Mesa Boogie Badlander 6L6
29	CT-BOG2CP-CL	Mesa Boogie Mark IIC+
30	CT-BOGF30-CL	Mesa Boogie F-30
31	CT-BRIT-M4-DS	Marshall JVM / Mode 4
32	CT-BZERO-OD	Mezzabarba Mzero
33	CT-CHUGGA-DS	Fortin Meshuggah 2.0
34	CT-CPLEX-OD	Cornell Plexi Ltd
35	CT-ED50-DS	EVH 5150III 50W 6L6
36	CT-EGNAT-CL	Egnater Tourmaster 4100
37	CT-FJOSE-DS	Friedman Jose
38	CT-IRAIL-DS	Engl Iron Ball 20
39	CT-JIMI-2-OD	PRS HDRX100 (Hendrix Woodstock-era)
40	CT-MULTI-DS	Koch Multitone
41	CT-PT50-DS	Custom Audio Amplifiers PT-50 (Pete Thorn Signature)
42	CT-TRILY-DS	Hughes & Kettner Trilogy
43	CT-XM100-DS	Masotti X100M Classic EL34
44	M-D102-BR-CR	Dumble 102 DRIVE_BRIGHT_C
45	M-D102-BR-OD	Dumble 102 DRIVE_BRIGHT_S
46	M-E570-CL	ENGL E570
47	M-E570-DS	ENGL E570
48	M-E530-MO-DS	ENGL E530
49	M-EV51-808-DS	EVH 5150III 100W 6L6GC 808
50	M-EV51-kuro-DS	EVH 5150III 100W 6L6GC kuro brighter
51	M-EV51-SD1-DS	EVH 5150III 100W 6L6GC sd-1

AMP

No.	Name	Real Amp/Pedal	No.	Name	Real Amp/Pedal
52	M-HOROD-BR-CL-FG	FENDER Hot Rod Deluxe clean FULL-RIG MIX	70	M-OR-AD30-CH1-2	Orange AD30 Ch2 with U87
53	M-HOROD-CR-FG	FENDER Hot Rod Deluxe Drive FULL-RIG MIX	71	M-EV-51-III-FG	EVH 5150 III MIX
54	M-HOROD-MD-OD-FG	FENDER Hot Rod Deluxe Drive FULL-RIG MIX	72	M-AC30-TB-FG	AC30TB MIX
55	M-HOROD-OD-FG	FENDER Hot Rod Deluxe More Drive FULL-RIG MIX	73	M-AC30-TB-HI-FG	AC30TB higher-gain MIX
56	M-MS-GV2-DS	Marshall Guv'hor 2 PEDAL	74	M-AC30-TB-HOT-FG	AC30TB Ch2 Hot On V6 MIX
57	M-MS-BLUESB-DS	Marshall Bluesbreaker MkII	75	M-AC30-TB-HOT-FUL-L-FG	AC30TB Ch2 Hot On V10 MIX
58	M-MS-J9-OD	JCM 900 Hi-Gain Dual Reverb	76	M-ME-DR-Chu-FG	Mesa/Boogie Dual Rectifier(Chupacabra) U87
59	M-MS-J9-808-DS	JCM 900 Hi-Gain Dual Reverb with 808	77	M-ME-DR-MW-1-CH-2-DT	Mesa/Boogie Dual Rectifier Multi-Watt CH2 With Fortin Dirty Tree
60	M-MS-410-CL1-FG	JVM410H CH1 G75 with U87	78	M-ME-DR-MW-2-CH-3-TR	Mesa/Boogie Dual Rectifier Multi-Watt CH3 With Triumph(Precision Drive)
61	M-MS-410-CL2-FG	JVM410H CH1 G85 with U87	79	M-MT100-LEAD	PRS MT 100
62	M-MS-410-OD1-FG	JVM410H CH3 G80 with U87	80	M-PV605-LD-FG	Peavey 6505 II Lead, with Marshall 1960BV
63	M-MS-410-OD2-FG	JVM410H CH3 G75 with U87	81	M-RIV-SD1	Rivera Amplifiers
64	M-ME-DR-3CH	Mesa/Boogie Dual Rectifier Solo Head	82	M-RIV-OS2	Rivera Amplifiers
65	M-ME-DR-3CH_808	Mesa/Boogie Dual Rectifier Solo Head With 808	83	M-PV605-LD2-FG	Peavey 6505 II Lead, with Diezel 4x12 FL V30
66	M-ME-DR-3CH_TS9	Mesa/Boogie Dual Rectifier Solo Head With TS9	84	M-PVU-CH3	Peavey Ultra CH3
67	M-SLO100-NRM-CRN-FG	Soldano SLO-100 CLEAN mix	85	M-LANY-IRT60	Laney IRT60
68	M-SLO100-OD	Soldano SLO-100 OD	86	M-LANY-IRT60-LD	Laney IRT60
69	M-OR-AD30-CH2-1	Orange AD30 Ch2 with U87	87	M-HT5-CL	Blackstar HT5
			88	M-HT5-OD	Blackstar HT5
			89	M-HT5-OD-FG	Blackstar HT5 With sm57
			90	M-HT5-CL-FG	Blackstar HT5 With sm57

CAB

No.	Name	Cabinet	Speaker	Microphone(s)
1	RonaldJC120.CG_87_0in_CapEdge	Roland JC-120	Roland Built-in 12"	Neumann U87
2	Match412GAP.CG_7B_0in_CapEdge	Matchless 4x12	Goodmans Audiom 12P	Shure SM7B
3	BognaB212VTY.CG_58_0in_CapEdge	Bogner 2x12	Celestion V-Type	Shure SM58
4	Hess212VTY.CG_I5_57_Mix1	Hesu 2x12	Celestion V-Type	Audix i5 + Shure SM57 mix
5	Mgan112C90.CG_414s_2in_CapEdge	Morgan 1x12	Celestion Custom 90	AKG C414 XLS
6	HK412PJ_NV_121_2in_Cap	Hughes & Kettner 4x12	Celestion G12-PJ	Royer R121
7	Fend312FBL_160_409_Mix1	Fender 1x12	Fender	Beyerdynamic M160 + Sennheiser MD409 mix
8	FriedV212V30_421_7B_Mix1	Friedman 2x12	Celestion Vintage 30	Sennheiser MD421 + Shure SM7B mix
9	Eddi3412V30.CG_609_421_Mix2wav	EVH 5150III 4x12	Celestion Vintage 30	Sennheiser e609 + Sennheiser MD421 mix
10	BogOS412V30_1073_57_CapEdgeOAEQ	Mesa Boogie OS 4x12	Celestion Vintage 30	Shure SM57
11	VTH412V30E_NV_57_121_Mix1 1	VHT / Fryette 4x12	Celestion Vintage 30	Shure SM57 + Royer R121 mix
12	PivyCS212T75.CG_7B_0in_Cap	Peavey Classic 2x12	Celestion G12T-75	Shure SM7B
13	Gas412V30.CG_57_0in_CapEdge	Diezel 4x12	Celestion Vintage 30	Shure SM57
14	Brit1960BB_58_121_Mix1	Marshall 1960B 4x12	Celestion G12T-75 / V30	Shure SM58 + Royer R121 mix
15	FDTWRFBF_NV_84_0in_Cap	Fender Twin Reverb	Fender	Neumann KM84
16	Marco102_NV_121_2in_Cap	Markbass CMD 1x12	Markbass Custom	Royer R121
17	MarcoNY212_NV_409_0in_Cap	Markbass New York 2x12	Markbass Custom	Sennheiser MD409
18	BogieWay210_NV_609_0in_Cap	Mesa Boogie Walkabout 2x10	Eminence Walkabout speaker	Sennheiser e609
19	AmpegSVT410.CG_57_0in_Cap	Ampeg SVT 4x10	Ampeg/Eminence SVT speaker	Shure SM57
20	Angel212V30E.CG_7_121_Mix1	ENGL 2x12	Celestion Vintage 30	Shure SM7B + Royer R121 mix
21	Angel412V30.CG_160_0_Cap	ENGL 4x12	Celestion Vintage 30	Beyerdynamic M160
22	BadBLK112SWT.CG_160_0in_Cap	Bad Cat Black Cat 1x12	Celestion Sidewinder	Beyerdynamic M160
23	Barber212V30E_57_7B_Mix1	Barber 2x12	Celestion Vintage 30	Shure SM57 + Shure SM7B mix

CAB

No.	Name	Cabinet	Speaker	Microphone(s)
24	Bogie412MV_160_121_Mix1	Mesa Boogie 4x12	Celestion Vintage 30	Beyerdynamic M160 + Royer R121 mix
25	BognaGF212T75.CG_103_0in_Cap	Bogner Goldfinger 2x12	Celestion G12T-75	Neumann TLM103
26	BognaGF212T75.CG_609_0in_CapEdge	Bogner Goldfinger 2x12	Celestion G12T-75	Sennheiser e609
27	BogOS412V30_1073_121_0_CapEQ	Mesa Boogie OS 4x12	Celestion Vintage 30	Royer R121
28	Brit1960BB_7B_121_Mix1	Marshall 1960B 4x12	Celestion G12T-75 / V30	Shure SM7B + Royer R121 mix
29	Brit425TEX.CG_609_0in_Cap	Marshall 425 4x12	Eminence Texas Heat	Sennheiser e609
30	Brit425TEX.CG_609_2in_CapEdge	Marshall 425 4x12	Eminence Texas Heat	Sennheiser e609
31	Bstar412MAN.CG_160_0in_Cap	Blackstar 4x12	Celestion Man O' War	Beyerdynamic M160
32	CarVai212ATY.CG_7B_0in_CapEdge	Carvin Legacy (Vai) 2x12	Celestion A-Type	Shure SM7B
33	DzB212G12ME.CG_160_2in_CapEdge	Diezel 2x12	Celestion G12M-65 Creamback	Beyerdynamic M160
34	Edd212V30E_160_57_Mix1	EVH 2x12	Celestion Vintage 30	Beyerdynamic M160 + Shure SM57 mix
35	EddIC412G65.CG_906_0in_CapEdge	EVH Iconic 4x12	Celestion G12-65	Sennheiser e906
36	Fend59BBB.NV_414s_2in_Cap	Fender '59 Bassman 4x10	Jensen P10R	AKG C414 XLS
37	Gampe_SVT810.NV_160_0in_Cap	Ampeg SVT 8x10	Eminence SVT speaker	Beyerdynamic M160
38	Haide412V30.CG_160_121_Mix1	Hayden 4x12	Celestion Vintage 30	Beyerdynamic M160 + Royer R121 mix
39	Hess212AEDemon.CG_160_0_Cap	Hesu 2x12	Hesu Demon	Beyerdynamic M160
40	Hess412CB65.CG_58_0in_CapEdge	Hesu 4x12	Celestion Creamback G12M-65	Shure SM58
41	HK412G12ME_7B_421_Mix1	Hughes & Kettner 4x12	Celestion G12M-65 Creamback	Shure SM7B + Sennheiser MD421 mix
42	Match412AlnGld.CG_906_0in_CapEdge	Matchless 4x12	Celestion Alnico Gold	Sennheiser e906
43	Mgan112AlBlu_7B_121_Mix1	Morgan 1x12	Celestion Alnico Blue	Shure SM7B + Royer R121 mix
44	Ornage412G12M.NV_160_0in_Cap	Orange PPC 4x12	Celestion G12M Greenback	Beyerdynamic M160
45	PivyCS212M20.CG_121_2in_Cap	Peavey Classic 2x12	Celestion G12M-20 Greenback	Royer R121
46	SuperDK12.NV_414s_0in_Cap	Supro Delta King 1x12	Supro DK-12	AKG C414 XLS
47	VocsAC15AlSlv.CG_7_160_Mix1	Vox AC15 1x12	Celestion Alnico Silver	Shure SM7B + Beyerdynamic M160 mix
48	WME212AlnGld.CG_57160_0in_CapEdge	Wampler / Custom 2x12	Celestion Alnico Gold	Shure SM57 + Beyerdynamic M160 mix
49	WME212CB75.CG_906_0in_CapEdge	Wampler / Custom 2x12	Celestion Creamback G12H-75	Sennheiser e906
50	ME_RO_KING_412_MIX	MESA ROAD KING CLOSED BACK 4x12	Celestion Vintage 30	Shure SM7B + Beyerdynamic M160 mix
51	Mars_1960B_412_MIX	Marshall1960B	Celestion Creamback G12M-65	Beyerdynamic M160 + Sennheiser MD409 mix
52	ENGL XXL 4x12_57	ENGL XXL 4x12	Celestion Vintage 30	SM57
53	Fried_2x12_R121	Friedman 2x12	Celestion V-Type	Royer R-121
54	Fend_TWR_212_MIX	FenderTwinReverb	JensenC12K	Beyerdynamic M160 + Royer R121 mix
55	Peavey_4x12_EDGE_57_5	Peavey Classic 4x12 MID 5 cm 45°	Celestion Creamback G12M-65	Shure SM57
56	V0_AC_30_212_C414	VOXAC30	CelestionAlnicoSilverBelldrivers212	AKG C414 XLS
57	Hiwatt_4x12_1	Hiwatt 4x12	Fane Simms Watts	Sennheiser MD421 + Royer R-121
58	Hiwatt_4x12_2	Hiwatt 4x12	Fane Simms Watts	MD421 + Beyerdynamic M160
59	Me_TraditionalS_412_409	Mesa TraditionalStraight	Celestion Vintage 30	Sennheiser MD409
60	Mar410H_1960_U87	Marshall1960A	Celestion G12T-75	Neumann U87
61	Orang_PPC412_e906	Orange PPC412	Celestion Vintage 30	Sennheiser e906
62	Orang2_PPC412_M160	Orange PPC412	Celestion Vintage 30	Beyerdynamic M160
63	Suhr_212_MIX	Suhr 212	Celestion Vintage 30	Beyerdynamic M160 + Royer R121 mix
64	DiezelFL_412_D440	DiezelFL	Celestion K100	AKG D440
65	TraceElliot_4x12_57	TraceElliot 4x12	TraceElliot Speakers	Shure SM57
66	Bogner_Ubercab412_MIX	Bogner Ubercab 412	Celestion Creamback G12H-75	Shure SM57 + Sennheiser MD421
67	EVH_Straight_412_C414	EVH Straight 412	Celestion G12T-75	AKG C414 XLS
68	Peavey1_4x12_EDGE_57_30	Peavey Classic 4x12 EDGE 30cm	Celestion Creamback G12H-M65	Shure SM57
69	Peavey2_4x12_EDGE_57_2	Peavey Classic 4x12 EDGE 2.5cm 45°	Celestion Creamback G12H-M65	Shure SM57
70	Fried_2x12_e906	Friedman 2x12	Celestion Vintage 30	Sennheiser e906
71	Mars_1960B_412_M160	Marshall 1960B 4x12	Celestion G12T-75	Beyerdynamic M160

CAB

No.	Name	Cabinet	Speaker	Microphone(s)
72	Mars_1935B_MIX	Marshall 1935B	Celestion G12T-75+Celestion G10T-25	Beyerdynamic M160 + Sennheiser MD409 mix
73	V0_AC_30_212_57	Vox AC30 212	Celestion "Blue" Alnico	Shure SM57
74	Fend_TWR_212_MIX2	Fender Twin Reverb '65 Reissue 2x12	Jensen C12K	Shure SM7B + Sennheiser e609
75	Fend_BsmanTw_4x10_MIX	Fender Bassman Tweed 4x10	Jensen P10R	Shure SM7B + Royer R121 mix
76	Boogie_Standard_4x12_D440	Mesa/Boogie Rectifier Standard 4x12	Celestion Vintage 30	AKG D440
77	Bogner_212_D440	Bogner 2x12	Electro-Voice EVM-12L	Shure SM7B
78	Hartke410_SM7B	Hartke 410	Hartke Speakers	Sennheiser e906
79	EVH_412_Angled_MIX	EVH 412 Angled	Celestion EVH 20W	Beyerdynamic M160 + Sennheiser MD409 mix
80	LANEY_Bs100_4x12_R121	LANEY Bass 100 4x12	Goodmans 241	Royer R-121
81	Peavey_Classic30_E906b	Peavey Classic 30	Celestion "Blue" Alnico	Sennheiser e906B
82	Mars_1960A+B_MIX	Marshall 1960A+B	Celestion G12T-75	Shure SM57 + Shure SM7B mix
83	Diezel_212_57	Synergy-Diezel 212	Celestion Vintage 30	Shure SM57
84	Wangs112_U87	Biyang Wangs 112 open	Celestion G12T-75	Neumann U87
85	V0_TB18C1_57	Vox TB18C1	Vox 110 Speakers	Shure SM57
86	H&K_4x12_e609	Hughes & Kettner 4x12	Celestion G12M Greenback	Sennheiser e609
87	H&K_4x12_MIX	Hughes & Kettner 4x12	Celestion G12M Greenback	Beyerdynamic M160 + Sennheiser MD409 mix
88	HiwattSE412_MIX	Hiwatt SE412	Celestion G12-65	Shure SM7B + Sennheiser e609
89	Ampeg4x12_FLAT58	Ampeg 4x12	Celestion Vintage 30	Flat Shure SM58
90	DarkglassDG212N_C414	Darkglass DG212N	Custom Eminence Neody Speakers	AKG C414 XLS

Statement:

The names of manufacturers and products mentioned in this list are solely used to objectively describe the timbre performance and functional characteristics of the relevant products. They do not constitute a recommendation, recognition, or endorsement of any manufacturer or product, nor do they involve any form of quality guarantee or usage advice.

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TANK PRO

Effects Description

DRUM (100)

Effect Name	Feature Description
DRUM (100)	Equipped with 100 diverse rhythm patterns spanning a wide range of musical styles including pop, rock, jazz, folk, Latin, and more, catering to performance and creative needs across various scenarios. Supports flexible fill-in functionality, enabling instant rhythmic embellishments to enhance melodic layering and live expressiveness — ideal for improvisation, arrangement, and music production.

WAH (14)

Effect Name	Feature Description	Parameter Adjustment
X-Wah	Wah pedal effect voiced after the VOX Clyde McCoy wah pedal.	Value: Adjusts the combined value of the core wah frequency and frequency range
Funk-Wah	Wah pedal effect voiced after the Xotic XW-2 wah pedal.	Gain: Adjusts the gain amount of the wah effect Level: Adjusts the output volume of the wah effect
Slide-Wah	Wah pedal effect voiced after the VOX V846 wah pedal.	
Cry-Wah	Wah pedal effect voiced after the Dunlop Cry Baby wah pedal.	
Wah-Wah	Auto-wah effect that produces a rhythmic, undulating tonal movement based on a user-defined frequency parameter.	Speed: Adjusts the rate of the wah effect Q: Adjusts the ratio of center frequency to waveform bandwidth Width: Adjusts the overall waveform width parameter ratio Mix: Adjusts the overall waveform width parameter ratio Level: Adjusts the output volume of the wah effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Sense-Wah	Dynamic-sensing auto-wah effect that automatically triggers and adjusts the wah response according to the playing dynamics of the instrument.	Sense: Adjusts the sensitivity Attack: Adjusts the onset time after the effect is triggered Q: Adjusts the ratio of center frequency to waveform bandwidth fPeak: Adjusts the feedback depth of the sweep effect in the wet signal Mix: Adjusts the overall waveform width parameter ratio Width: Adjusts the overall waveform width parameter ratio Level: Adjusts the output volume of the wah effect
Compress	Compression effect for dynamic control.	Sustain: Adjusts the combined compression threshold and ratio Attack: Adjusts the compression attack speed Level: Adjusts the output volume after compression Blend: Adjusts the mix ratio between the compressed and dry signals
Compress Pro	Professional-grade compression effect with multiple adjustable parameters.	Ratio: Adjusts the compression ratio Gain: Adjusts the input gain Knee: Adjusts the transition smoothness of the compression onset Thd: Adjusts the threshold at which compression is triggered Attack: Adjusts the compression attack speed Level: Adjusts the output volume after compression Blend: Adjusts the mix ratio between the compressed and dry signals
F Compress	Compression effect combining dynamic control with adjustable post-compression brightness.	Ratio: Adjusts the compression ratio Gain: Adjusts the input gain Knee: Adjusts the transition smoothness of the compression onset Thd: Adjusts the threshold at which compression is triggered Attack: Adjusts the compression attack speed Tone: Adjusts the brightness of the compression effect Level: Adjusts the output volume after compression Blend: Adjusts the mix ratio between the compressed and dry signals
Boost	Enhancer effect centered on boosting signal strength.	Gain: Adjusts the input gain Level: Adjusts the output volume after the enhancer effect
A Boost	Type-A enhancer effect.	Gain: Adjusts the input gain
B Boost	Type-B enhancer effect.	Bass: Adjusts the low-frequency response of the enhancer effect
E Boost	Type-E enhancer effect.	Mid: Adjusts the mid-frequency response of the enhancer effect Treble: Adjusts the high-frequency response of the enhancer effect Level: Adjusts the output volume after the enhancer effect
Boost ED	Overdrive-type enhancer effect with integrated overdrive functionality.	Gain: Adjusts the input gain Grit: Adjusts the overdrive saturation of the enhancer effect Level: Adjusts the output volume after the enhancer effect

FX1&FX2 (20)

Effect Name	Feature Description	Parameter Adjustment
Noise Gate++	Noise reduction effect; an AI-driven signal processing tool that controls signal on/off switching.	Gate: Adjusts the noise gate trigger threshold
AI Gate MS	Noise reduction effect based on a statistical deviation tracking model with intelligent algorithm processing.	Gate: Adjusts the noise gate trigger threshold Bias: Adjusts the pre-activation lead time for noise reduction preparation
Soft Gate	Noise reduction effect that monitors and controls audio levels to reduce noise, enhance clarity, and improve overall audio quality.	Thd: Adjusts the noise reduction threshold
Hard Gate	Noise reduction effect with a more aggressive processing approach than Soft Gate when handling audio signals.	Thd: Adjusts the noise reduction threshold
Pro Gate	Noise reduction effect that sets a precise threshold to identify and eliminate low-level background noise, optimizing audio purity.	Att: Adjusts the time for the noise gate to fully open after the signal exceeds the threshold Rel: Adjusts the time for the noise gate to close after the signal falls below the threshold Thd: Adjusts the noise reduction threshold Kw: Adjusts the width of the transition zone above and below the threshold Ratio: Adjusts the ratio between input noise and output signal
Compress	Compression effect for dynamic control.	Sustain: Adjusts the combined compression threshold and ratio Attack: Adjusts the compression attack speed Level: Adjusts the output volume after compression Blend: Adjusts the mix ratio between the compressed and dry signals
Compress Pro	Professional-grade compression effect with multiple adjustable parameters.	Ratio: Adjusts the compression ratio Gain: Adjusts the input gain Knee: Adjusts the transition smoothness of the compression onset Thd: Adjusts the threshold at which compression is triggered Attack: Adjusts the compression attack speed Level: Adjusts the output volume after compression Blend: Adjusts the mix ratio between the compressed and dry signals
F Compress	Compression effect combining dynamic control with adjustable post-compression brightness.	Ratio: Adjusts the compression ratio Gain: Adjusts the input gain Knee: Adjusts the transition smoothness of the compression onset Thd: Adjusts the threshold at which compression is triggered Attack: Adjusts the compression attack speed Tone: Adjusts the brightness of the compression effect Level: Adjusts the output volume after compression Blend: Adjusts the mix ratio between the compressed and dry signals
Boost	Enhancer effect centered on boosting signal strength.	Gain: Adjusts the input gain Level: Adjusts the output volume after the enhancer effect
A Boost	Type-A enhancer effect.	Gain: Adjusts the input gain
B Boost	Type-B enhancer effect.	Bass: Adjusts the low-frequency response of the enhancer effect
E Boost	Type-E enhancer effect.	Mid: Adjusts the mid-frequency response of the enhancer effect Treble: Adjusts the high-frequency response of the enhancer effect Level: Adjusts the output volume after the enhancer effect
Boost ED	Overdrive-type enhancer effect with integrated overdrive functionality.	Gain: Adjusts the input gain Grit: Adjusts the overdrive saturation of the enhancer effect Level: Adjusts the output volume after the enhancer effect
Lofi	Lo-Fi effect that emulates the sonic characteristics of vintage playback equipment.	Bit: Adjusts the bit rate Level: Adjusts the effect output volume Filter: Adjusts a specific frequency band of the filter

FX1&FX2 (20)

Effect Name	Feature Description	Parameter Adjustment
Pitch	Pitch shifting effect that alters the pitch of an audio signal by modifying its frequency.	HPitch: Adjusts the pitch of the high-pitch component LPitch: Adjusts the pitch of the low-pitch component HighLvl: Adjusts the output volume of the high-pitch component LowLvl: Adjusts the output volume of the low-pitch component DryLvl: Adjusts the output volume of the original dry signal
Pitch shifter	Pitch shift effect with a range of ± 12 semitones.	Semi: Pitch offset amount
Octave	Octave effect that adds a pitch one octave above or one octave below the original signal.	HighLvl: Adjusts the output volume of the "one octave up" signal LowLvl: Adjusts the output volume of the "one octave down" signal DryLvl: Adjusts the output volume of the original dry signal
Ring	Simulated bell effect that recreates the resonant characteristics of bells and chimes.	Freq: Adjusts the frequency of the bell effect Mix: Adjusts the blend ratio between the bell effect and the original signal
Exp_Vol	Volume control effect for an external expression pedal; recommended for use with the Toe position set to None in the Pedal settings.	Vol: Adjusts the output volume (assign the Vol parameter to an expression pedal for use)
Exp_Pan	Left/right channel output balance control effect for an external expression pedal; recommended for use with the Toe position set to Resume in the Pedal settings.	Pan: Adjusts the left/right channel output balance (assign the Pan parameter to an expression pedal for use)

DS (30)

Naming	Prototype	Parameter Adjustment
See attached tone list	See attached tone list	Gain: Adjusts the gain of the module Level: Adjusts the output volume of the module Bass: Adjusts the low-frequency response of the module Middle: Adjusts the mid-frequency response of the module Treble: Adjusts the high-frequency response of the module Reso: Adjusts the low-frequency resonance of the module Pres: Adjusts the presence of the module Bright: Adjusts the brightness of the module

AMP (90)

Naming	Prototype	Parameter Adjustment
See attached tone list	See attached tone list	Gain: Adjusts the gain of the module Level: Adjusts the output volume of the module Bass: Adjusts the low-frequency response of the module Middle: Adjusts the mid-frequency response of the module Treble: Adjusts the high-frequency response of the module Reso: Adjusts the low-frequency resonance of the module Pres: Adjusts the presence of the module Bright: Adjusts the brightness of the module

CAB (90)

Naming	Prototype	Parameter Adjustment
See attached tone list	See attached tone list	Level: Adjusts the output volume of the cabinet Lcut: Adjusts the low-cut frequency point of the cabinet Hcut: Adjusts the high-cut frequency point of the cabinet

EQ (8)

Effect Name	Feature Description	Parameter Adjustment
Guitar EQ 6	6-band guitar equalizer for precise tonal adjustment across different frequency ranges of the guitar.	100Hz: Adjusts the loudness of frequencies around 100 Hz 200Hz: Adjusts the loudness of frequencies around 200 Hz 400Hz: Adjusts the loudness of frequencies around 400 Hz 800Hz: Adjusts the loudness of frequencies around 800 Hz 1.6kHz: Adjusts the loudness of frequencies around 1,600 Hz 3.2kHz: Adjusts the loudness of frequencies around 3,200 Hz
Bass EQ 7	7-band bass equalizer for precise tonal adjustment across different frequency ranges of the bass.	50Hz: Adjusts the loudness of frequencies around 50 Hz 120Hz: Adjusts the loudness of frequencies around 120 Hz 400Hz: Adjusts the loudness of frequencies around 400 Hz 500Hz: Adjusts the loudness of frequencies around 500 Hz 800Hz: Adjusts the loudness of frequencies around 800 Hz 4.5kHz: Adjusts the loudness of frequencies around 4,500 Hz 10kHz: Adjusts the loudness of frequencies around 10,000 Hz
Normal EQ 10	10-band equalizer for precise tonal adjustment spanning from sub-bass to ultra-high frequencies.	31.25Hz: Adjusts the loudness of frequencies around 31.5 Hz 62.5Hz: Adjusts the loudness of frequencies around 62.5 Hz 125Hz: Adjusts the loudness of frequencies around 125 Hz 250Hz: Adjusts the loudness of frequencies around 250 Hz 500Hz: Adjusts the loudness of frequencies around 500 Hz 1kHz: Adjusts the loudness of frequencies around 1,000 Hz 2kHz: Adjusts the loudness of frequencies around 2,000 Hz 4kHz: Adjusts the loudness of frequencies around 4,000 Hz 8kHz: Adjusts the loudness of frequencies around 8,000 Hz 16kHz: Adjusts the loudness of frequencies around 16,000 Hz
Lofi	Lo-Fi effect that emulates the sonic characteristics of vintage playback equipment.	Bit: Adjusts the bit rate Level: Adjusts the effect output volume Filter: Adjusts a specific frequency band of the filter
Pitch	Pitch shifting effect that alters the pitch of an audio signal by modifying its frequency.	HPitch: Adjusts the pitch of the high-pitch component LPitch: Adjusts the pitch of the low-pitch component HighLvl: Adjusts the output volume of the high-pitch component LowLvl: Adjusts the output volume of the low-pitch component DryLvl: Adjusts the output volume of the original dry signal
Pitch shifter	Pitch shift effect with a range of ± 12 semitones.	Semi: Pitch offset amount
Octave	Octave effect that adds a pitch one octave above or one octave below the original signal.	HighLvl: Adjusts the output volume of the "one octave up" signal LowLvl: Adjusts the output volume of the "one octave down" signal DryLvl: Adjusts the output volume of the original dry signal
Ring	Simulated bell effect that recreates the resonant characteristics of bells and chimes.	Freq: Adjusts the frequency of the bell effect Mix: Adjusts the blend ratio between the bell effect and the original signal

MOD (20)

Effect Name	Feature Description	Parameter Adjustment
Chorus	Chorus effect that simulates the sound of multiple vocalists or instruments performing together, adding fullness, spatial depth, and thickness to a single tone — a widely used tool for enhancing tonal expressiveness.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Tri Chorus	Triple chorus effect — an advanced form of standard chorus that generates 3 independent sub-signals to simulate a more natural and full-bodied multi-voice/multi-instrument sound, delivering greater layering and more refined spatial dimension than conventional chorus.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Flanger	Flanger effect that blends a delayed signal with the original to produce a periodic sweeping sensation (reminiscent of the rippling sound of a passing aircraft), imparting a strong sense of dynamic movement and spatial modulation — commonly used with electric guitar, synthesizers, and similar instruments.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Fb: Adjusts the feedback amount of the sweep effect in the wet signal Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Tri Flanger	Triple flanger effect — an advanced form of standard flanger that stacks three independent flanging signals to generate a more complex and finely layered sweeping modulation. The result is a richer, more intricate sound (akin to multiple flanged instruments sounding simultaneously), well-suited for scenarios that demand intense dynamics without monotony.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Fb: Adjusts the feedback amount of the sweep effect in the wet signal Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Tremolo	Tremolo effect that periodically modulates the output volume (without altering pitch, distinct from vibrato) to produce a breathing or pulsing sensation, similar to the sound of rapidly and manually adjusting the volume.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Level: Adjusts the output volume of the module Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Tri Tremolo	Triple tremolo effect — an advanced take on traditional tremolo that replaces a single regular rhythm with three interwoven rhythmic patterns, creating a richer, more layered volume modulation to avoid monotony and elevate the dynamic complexity of the tone.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Level: Adjusts the output volume of the module Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Opto Tremolo	Optical tremolo that approximates the tremolo circuit of vintage Fender amplifiers, delivering a more defined and crisp character. Tremolo intensity responds to playing dynamics, with clear articulation, pronounced effect on high frequencies, a subtle Vibe-like quality, and a traditional, organic feel — ideal for adding color and creating a distinctive musical atmosphere.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Level: Adjusts the output volume of the module Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Phaser	Phaser effect that shifts the phase of an audio signal and blends it with the original to produce an ethereal, psychedelic sweeping sound, suitable for guitar processing and film/media audio production. Its comb filter notches are non-harmonically related, resulting in a smooth, gentle tone without the sharp metallic edge characteristic of flanging.	Speed: Adjusts the rate of the effect MidCut: Adjusts the intensity of phase shifting in the mid-frequency range Reso: Adjusts the resonance amount within a specific frequency range Fb: Adjusts the feedback amount of the sweep effect in the wet signal Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Vibrato	Vibrato effect that generates modulation through periodic pitch fluctuation — its defining characteristic being pitch variation, as opposed to the volume variation of tremolo — widely used in instrument performance and audio processing. At its core, it produces a regular upward and downward deviation of pitch around a center reference, similar to the natural pitch undulation of the human singing voice.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Tri Vibrato	Triple vibrato effect that applies three simultaneous or interleaved vibrato modulations — each with different rates, depths, or waveforms — to the same audio signal, enhancing tonal layering and expressiveness (e.g., low speed and shallow depth for gentle rhythm, mid speed and depth for added dynamics, high speed with adjustable depth for intensified undulation). Optical vibrato effect that modulates the	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Opto Vibrato	amplitude of an audio signal based on optical principles, producing natural, smooth, periodic volume variations with minimal distortion. It adds dynamic movement to the sound, making the tone fuller and more vibrant — commonly used to enhance instrument expressiveness (particularly guitar) and to create a warm, vintage sonic atmosphere.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Univibe	Uni-Vibe effect, originally designed to emulate a rotary speaker, that produces a distinctive "swirling" tone combining warmth and psychedelic character — an iconic tool for guitarists and music producers. By integrating phase modulation with dynamic filtering, it delivers a more "organic" sound with pronounced low-frequency harmonics and a natural "breathing" quality.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Tri Univibe	Triple Uni-Vibe effect, based on the Uni-Vibe principle, that employs three interacting oscillators to generate complex phase and amplitude modulation, producing a "swirling" sensation as if sound is expanding and contracting through space — delivering both a powerful sense of spatial depth and dynamic expressiveness.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Autofilter	Auto-filter effect — a dynamic audio processor that automatically controls filter parameters via preset or external trigger signals to achieve dynamic tonal variation, transcending the limitations of manual adjustment. Widely used in guitar, synthesizer, and electronic music production, it is a key tool for crafting "dynamic tones," bridging expressive playing, rhythmic arrangement, and sound design to bring greater vitality to the audio signal.	Speed: Adjusts the rate of the effect Min: Sets the minimum value of the sweep frequency range Max: Sets the maximum value of the sweep frequency range Mix: Adjusts the wet/dry blend ratio of the effect Fb: Adjusts the feedback amount of the sweep effect in the wet signal Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Phaser Stereo	Stereo phaser effect — an evolution of the traditional mono phaser that retains the core sweeping quality of phase modulation while applying differentiated processing to the left and right channels, delivering a stronger sense of spatial depth and dynamic movement. It is a key tool for creating a "flowing" and "spatial traversal" sensation. Its core advantage lies in the desynchronized phase variation between the left and right channels, elevating the sweep effect from a flat tonal shift to a fully three-dimensional spatial motion.	Speed: Adjusts the rate of the effect Stage: Selects the frequency processing mode. Mode A applies phase shifting to a single frequency point; Mode B applies phase shifting to multiple frequency points simultaneously. Regen: Adjusts the feedback intensity of the effect Mix: Adjusts the wet/dry blend ratio of the effect Lfo: Selects the LFO waveform, including Sine (Sin), Square (Sqw), Sawtooth (Saw), Triangle (Tri), and Custom Waveform (Wav) Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Flanger Stereo	Stereo flanger effect — an evolution of the traditional mono flanger that preserves the core "delay-stacked sweep" character (such as the metallic quality of a "passing aircraft" or the silky smoothness of "high-speed tape drag"), while applying differentiated processing to the left and right channels to elevate the effect from flat modulation to three-dimensional spatial motion. It is a key tool for creating a sense of "surround envelopment" and "dynamic spatial traversal."	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Regen: Adjusts the feedback intensity of the effect Mix: Adjusts the wet/dry blend ratio of the effect Jet: Adjusts the intensity of the jet-like sweeping sensation Lfo: Selects the LFO waveform, including Sine (Sin), Square (Sqw), Sawtooth (Saw), Triangle (Tri), and Custom Waveform (Wav) Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Vibe Stereo	Stereo Uni-Vibe effect that utilizes deep positive feedback to drive electronic components into alternating conduction and cutoff states, generating rectangular pulse waves rich in high-order harmonics, while applying differentiated modulation parameters to the left and right channels to create a stereo effect. Unlike the stereo flanger (which uses delay-stacked sweeping with a characteristic "jet" sound) and the stereo phaser (which modulates spectral peaks and troughs via all-pass filtering), this effect focuses on producing a rotating, undulating spatial sensation with a distinctive harmonic timbre, and is also capable of rendering diverse effects such as chorus.	Speed: Adjusts the rate of the effect Stage: Selects the frequency processing mode. Mode A applies phase shifting to a single frequency point; Mode B applies phase shifting to multiple frequency points simultaneously. Regen: Adjusts the feedback intensity of the effect Mix: Adjusts the wet/dry blend ratio of the effect Lfo: Selects the LFO waveform, including Sine (Sin), Square (Sqw), Sawtooth (Saw), Triangle (Tri), and Custom Waveform (Wav) Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Chorus Stereo	Stereo chorus effect — a widely adopted modulation audio effect whose core function is to simulate the natural variations of "multi-voice choral singing" or "multiple instruments performing in unison," creating rich spatial layering and tonal depth within the stereo field. It simultaneously broadens the sonic width and imparts a natural sense of movement, making it an essential tool for shaping the spatial dimension of instruments and vocals. It works by distributing processed "duplicate signals" across the left and right channels, introducing channel differences through phase, delay offset, or volume ratio adjustments, then blending the result with the original signal to produce the sensation of sound diffusing through a three-dimensional space.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Mode: Selects the modulation model, including Custom Harmonic (Harm), Sine (Sin), Square (Sqw), Sawtooth (Saw), Triangle (Tri), and Custom Waveform (Wav) Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Tremolo Stereo	Stereo tremolo effect — a modulation effect that shapes spatial dimension through differentiated periodic volume fluctuations between the left and right channels, enhancing rhythmic movement and breaking monotony. It is a commonly used tool for crafting the "breathing quality" of instruments and vocals. The differentiated fluctuations between channels produce a variety of listening experiences — such as a "left-to-right movement" when the left channel is louder than the right, or an "alternating left-right pulse" when the channels are in opposite phase — freeing the sound from a fixed spatial position.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Level: Adjusts the output volume of the module Mode: Selects the modulation model, including Sine (Sin), Tube-driven Current (Tube), and Optical Simulation (Phnt) Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Vibrato Stereo	Stereo vibrato effect — centered on enriching tone through periodic pitch fluctuation (as distinct from tremolo's volume modulation), with differentiated processing applied to the left and right channels to elevate the sound from "flat" to "three-dimensional," creating a stronger sense of spatial depth and layering. Its defining characteristic is the "spatialization of pitch fluctuation" — emphasizing neither thickness nor volume variation — making it a practical tool for enhancing tonal expressiveness.	Speed: Adjusts the rate of the effect Depth: Adjusts the depth of the effect Bits: Adjusts the quantization resolution of the digital signal processing Mix: Adjusts the wet/dry blend ratio of the effect Sync: When enabled, Speed is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Clean	Pure delay effect that generates natural, clean echoes with no complex modulation, preserving the purity of the original tone — ideal for straightforward, authentic sound requirements.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Modern	Modern delay effect that applies phase shifting to the delayed signal combined with LFO modulation, allowing the echoes to achieve both spatial depth and dynamic layering, avoiding monotony while delivering a rich and vibrant listening experience, all while retaining the clarity of the underlying tone.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Phaser: Adjusts the intensity of phase modulation applied to the delayed signal Mod: Adjusts the modulation intensity of the LFO (Low-Frequency Oscillator) on the all-pass filter Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Echo	Echo delay effect that simulates the reflection and attenuation of sound in physical space. Through the repetition and gradual volume decay of echoes, it imparts a sense of spatial depth and distinctive rhythmic layering, faithfully reproducing the physical characteristics of sound wave propagation.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Analog	Analog delay effect that employs DSP technology to faithfully recreate the tonal character and circuit behavior of traditional BBD chips, while maintaining the stability of digital processing. It reproduces the warm, lush, and organic classic texture of analog delay, with particular emphasis on natural high-frequency roll-off and the distinctive character of even-order harmonic distortion.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Duck	Ducking delay effect that dynamically monitors the dry signal level — automatically attenuating the delay wet signal when the dry signal is present (to prevent masking), and naturally restoring the wet signal during pauses or reductions in the dry signal. The Release parameter determines the speed of wet signal recovery: a short value produces an abrupt return, while a long value yields a smooth, gradual fade-in.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Release: Adjusts the speed at which the delayed signal volume returns to its normal level Speed: Adjusts the rate of the LFO (Low-Frequency Oscillator) Depth: Adjusts the depth of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Dtype	Tape delay effect that incorporates tape emulation processing, imparting a vintage, imperfect texture reminiscent of classic tape machines. The core parameter Grit controls the intensity of the processing (higher values produce greater signal degradation; at 0, only the LFO acts on the delay signal), while the LFO introduces subtle periodic pitch fluctuation.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Grit: Adjusts the intensity of the tape emulation processing applied to the delayed signal Speed: Adjusts the rate of the LFO (Low-Frequency Oscillator) Depth: Adjusts the depth of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Tremolo	Tremolo delay effect that superimposes periodic volume modulation onto the delayed signal, so that each echo carries both a time interval and a dynamic volume undulation, enhancing rhythmic movement and spatial layering. The tremolo cycle equals Time × Pattern — smaller Pattern values produce denser modulation, a value of 1/1 synchronizes the tremolo with the delay Time, and larger values produce more gradual, expansive undulation.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Pattern: Adjusts the note value of one tremolo cycle Speed: Adjusts the rate of the LFO (Low-Frequency Oscillator) Depth: Adjusts the depth of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Filter	Filter delay effect that applies filtering to the delayed signal, introducing tonal variation with each successive echo to create a more complex sense of space and tonal layering. The core parameter Filter controls the intensity of the filtering — higher values produce more pronounced processing, while a value of 0 results in no filtering.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Filter: Adjusts the intensity of the filter effect Speed: Adjusts the rate of the LFO (Low-Frequency Oscillator) Depth: Adjusts the depth of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (synced to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Dual	Dual delay effect that uses the T-Mode parameter to adjust the frequency ratio between two delay signals (time offset = Time × T-Mode), producing complex listening experiences such as double echoes and spatial layering. When T-Mode is less than 1/1, echoes cycle from weak to strong; at 1/1, the primary and secondary echoes overlap and repeat; above 1/1, echoes cycle from strong to weak.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect T-Mode: Adjusts the note value ratio between the primary and secondary delay signals Speed: Adjusts the rate of the LFO (Low-Frequency Oscillator) Depth: Adjusts the depth of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Lofi	Lo-Fi delay effect that combines delay processing with vintage distortion characteristics, using the Grit parameter to introduce saturation, distortion, and background noise, alongside LFO modulation to simulate the instability of aging equipment. The result is a nostalgic, raw audio texture well-suited to vintage musical atmospheres.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Grit: Adjusts the intensity of the vintage distortion processing applied to the delayed signal Speed: Adjusts the rate of the LFO (Low-Frequency Oscillator) Depth: Adjusts the depth of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Pattern	Pattern delay effect centered on preset rhythmic patterns, where delay echoes repeat according to a predefined rhythmic pattern (such as specific beat configurations or drum-based logic) rather than fixed time intervals. LFO modulation (Speed controls rate, Depth controls amplitude) introduces periodic variation into the rhythmic framework, preventing a mechanical feel.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Pattern: Selects the rhythmic pattern of the echoes (15 patterns available) Speed: Adjusts the rate of the LFO (Low-Frequency Oscillator) Depth: Adjusts the depth of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Ice	Ice delay effect that applies pitch adjustment to the delayed signal combined with dynamic LFO modulation, allowing the echoes to generate a unique harmonic texture and layering. This elevates the effect from simple repetition to an active compositional element within the musical structure, delivering both spatial depth and emotional tension.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Pitch: Adjusts the pitch of the delayed echo signal Mod: Adjusts the modulation intensity of the LFO (Low-Frequency Oscillator) Sync: When enabled, Time is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
Reverse	Reverse delay effect that time-reverses the delayed signal and applies phase shifting via a phaser, producing a distinctive sensation of sound traveling "backwards from the future to the present." The result combines a sense of novelty with mystery, enhancing the spatial dimension and three-dimensionality of the sound.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Phaser: Adjusts the blend ratio between the phase-shifted signal and the original reversed signal Mod: Adjusts the modulation intensity of the LFO (Low-Frequency Oscillator) within the phaser Sync: When enabled, Time is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM
PingPong Stereo	Stereo ping-pong delay effect that alternately triggers delay signals between the left and right channels, simulating the dynamic trajectory of sound "bouncing left and right." It exploits inter-channel phase and time differences to create a strong sense of spatial movement. Unlike conventional stereo delay with independent echoes, the signals here are causally linked, resulting in more pronounced fluidity.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the wet/dry blend ratio of the delay effect Sync: When enabled, Time is displayed in beat divisions (sync'd to the preset BPM) S-Bpm: Sync tempo, aligned with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Clean Stereo	Stereo pure delay effect that generates natural, clean echoes with no complex modulation, preserving the purity of the original tone — ideal for straightforward, authentic sound requirements. Inter-channel time and spatial differences enhance the sense of width and envelopment, creating a more expansive stereo soundscape.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Modern Stereo	Stereo modern delay effect that applies phase shifting to the delayed signal combined with LFO modulation, allowing echoes to achieve both spatial depth and dynamic layering — avoiding monotony while delivering a rich and vibrant listening experience, all while retaining the clarity of the underlying tone. Inter-channel time and spatial differences enhance the sense of width and envelopment, creating a more expansive stereo soundscape.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Phaser: Adjusts the phase modulation intensity of the delayed signal Mod: Adjusts the modulation depth of the Low-Frequency Oscillator (LFO) on the all-pass filter Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Echo Stereo	Stereo echo delay effect that simulates the reflection and attenuation of sound in physical space. Through the repetition and gradual volume decay of echoes, it imparts a sense of spatial depth and distinctive rhythmic layering, faithfully reproducing the physical characteristics of sound wave propagation. Inter-channel time and spatial differences enhance the sense of width and envelopment, creating a more expansive stereo soundscape.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Analog Stereo	Stereo analog delay effect that employs DSP technology to faithfully recreate the tonal character and circuit behavior of traditional BBD chips, while maintaining the stability of digital processing. It reproduces the warm, lush, and organic classic texture of analog delay, with particular emphasis on natural high-frequency roll-off and the distinctive character of even-order harmonic distortion. Inter-channel time and spatial differences enhance the sense of width and envelopment, creating a more expansive stereo soundscape.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Duck Stereo	Stereo ducking delay effect that dynamically monitors the dry signal level — automatically attenuating the delay wet signal when the dry signal is present (to prevent masking), and naturally restoring the wet signal during pauses or reductions in the dry signal. The Release parameter determines the speed of wet signal recovery: a short value produces an abrupt return, while a long value yields a smooth, gradual fade-in. Inter-channel time and spatial differences enhance the sense of width and envelopment, creating a more expansive stereo soundscape.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Release: Adjusts the rate at which the delayed signal volume returns to normal Speed: Adjusts the rate of change of the Low-Frequency Oscillator (LFO) Depth: Adjusts the modulation depth of the Low-Frequency Oscillator (LFO) Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Dtype Stereo	A stereo tape delay effect layered with tape simulation processing, imparting a vintage character and analog imperfection to the echoes. The core parameter Grit controls the processing intensity (higher values produce a more degraded sound; at 0, only the LFO acts on the delay signal), while the LFO introduces subtle periodic pitch fluctuations. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Grit: Adjusts the processing intensity of the tape simulation effect on the delayed signal Speed: Adjusts the rate of change of the Low-Frequency Oscillator (LFO) Depth: Adjusts the modulation depth of the Low-Frequency Oscillator (LFO) Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Tremolo Stereo	A stereo tremolo delay effect that applies periodic volume modulation to the delayed signal, allowing echoes to carry both rhythmic timing and dynamic amplitude variation for enhanced groove and spatial depth. The tremolo cycle = Time × Pattern; smaller Pattern values produce denser fluctuations, a value of 1/1 synchronizes with Time, and larger values yield more expansive undulations. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Pattern: Adjusts the note value of one tremolo cycle, affecting the density of modulation Speed: Adjusts the rate of change of the tremolo Depth: Adjusts the modulation depth of the tremolo Dive: Adjusts the dive depth of the tremolo Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Filter Stereo	A stereo filter delay effect that applies filter processing to the delayed signal, introducing tonal variation with each echo repeat to create a more complex sense of space and timbral layering. The core parameter Filter controls the filtering intensity — higher values produce more pronounced processing, while a value of 0 results in no filtering. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Filter: Adjusts the intensity of the filter effect Speed: Adjusts the rate of change of the tremolo Depth: Adjusts the modulation depth of the tremolo Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Dual Stereo	A stereo dual delay effect that uses the T-Mode parameter to adjust the frequency differential between two delay signals (time offset = Time × T-Mode), producing complex listening experiences such as double echoes and layered spatial depth. When T-Mode is less than 1/1, echoes cycle from soft to loud; at 1/1, primary and secondary echoes overlap and loop; above 1/1, echoes cycle from loud to soft. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal T-Mode: Adjusts the note value of the primary and secondary delay signals Speed: Adjusts the rate of change of the tremolo Depth: Adjusts the modulation depth of the tremolo Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Lofi Stereo	A stereo lo-fi delay effect that combines delay processing with vintage distortion characteristics. The Grit parameter introduces saturation, distortion, and background noise, while LFO modulation simulates the unstable fluctuations of analog hardware, delivering a nostalgic, raw audio texture ideally suited for retro musical aesthetics. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Grit: Adjusts the processing intensity of the vintage distortion effect on the delayed signal Speed: Adjusts the rate of change of the tremolo Depth: Adjusts the modulation depth of the tremolo Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Pattern Stereo	A stereo pattern delay effect centered on preset rhythmic patterns, where delay echoes repeat according to a predefined rhythmic structure (such as specific beat patterns or drum-based logic) rather than fixed time intervals. LFO modulation (Speed controls rate, Depth controls amplitude) adds periodic variation to the rhythmic framework, preventing a mechanical feel. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Pattern: Selects the rhythmic pattern of the echo repeats (15 patterns available) Speed: Adjusts the rate of change of the tremolo Depth: Adjusts the modulation depth of the tremolo Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Ice Stereo	A stereo ice delay effect that applies pitch shifting to the delayed signal, combined with dynamic LFO modulation, allowing echoes to generate a distinctive harmonic texture and sense of layering — elevating the effect from simple replication to an active compositional element within the musical structure, with both spatial depth and emotional resonance. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Pitch: Adjusts the pitch of the delay echoes Mod: Adjusts the modulation depth of the Low-Frequency Oscillator (LFO) Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM
Reverse Stereo	A stereo reverse delay effect that applies time-reversal processing to the delayed signal, followed by phase adjustment via a phaser, producing a distinctive sensation of sound traveling from the "future" back to the "present." This effect delivers a sense of novelty and mystery while enhancing the spatial depth and three-dimensionality of the sound. By leveraging timing and spatial differences between channels, it enhances the width and envelopment of the sound, creating a broader and more immersive stereo soundstage.	Time: Adjusts the time interval between the dry signal and the delayed signal Fb: Adjusts the number of delay echo repetitions Mix: Adjusts the ratio of wet delay signal to dry signal Phaser: Adjusts the blend ratio between the phase-shifted signal and the original reversed signal Mod: Adjusts the modulation depth of the Low-Frequency Oscillator (LFO) within the phaser Sync: When enabled, Time is displayed in beats (synchronized to the preset BPM) S-Bpm: Sync tempo, consistent with the preset BPM

Effect Name	Feature Description	Parameter Adjustment
Room	A room reverb effect that simulates the acoustic characteristics of a simple room: soft interior materials absorb a portion of the reflected sound, while the core signal reflects off the walls to form a natural reverb, faithfully reproducing the authentic sonic texture of a small space.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Depth: Adjusts the balance between early reflections and late reverb
Hall	A hall reverb effect that simulates the acoustic characteristics of large indoor spaces such as concert halls. The core sonic impression is one of expansive spatial depth with subtle diffusion — preserving detail clarity while naturally enveloping the dry signal to recreate a grand soundstage. Ideal for enhancing vocal transparency, extending the spatial presence of instruments (such as piano and strings), or imparting an authentic sense of "standing in a vast hall" to ambient sound and narration.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Depth: Adjusts the balance between early reflections and late reverb
Plate	A plate reverb effect that simulates the reflection of sound off a metal plate surface. As the audio signal reaches the plate, a portion is absorbed, a portion passes through, and the primary component reflects off the surface, generating the characteristic plate reverb response.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Mod: Adjusts the influence of the modulation signal on the carrier signal
Spring	A spring reverb effect representing one of the most iconic vintage reverb types in audio processing, with a distinctly mechanical character: the audio signal is transmitted into a spring assembly to induce resonance, and a dedicated transducer captures the harmonic feedback generated by the spring's vibration, ultimately simulating the sensation of being enclosed within a space fitted with spring-based acoustic structures.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Combs: Adjusts the intensity of the comb filter effect
Shimmer	A shimmer reverb effect in which the wet reverb signal incorporates pitch shifting, enabling creative atmospheric sound design — dissonant intervals evoke an eerie, unsettling backdrop, while harmonic intervals produce an ethereal, ethereally luminous quality, with reverb tails that bloom like the gentle glow of a rising dawn.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Tone: Adjusts the brightness of the wet reverb signal Pitch: Adjusts the pitch shift of the wet reverb signal Amount: Adjusts the intensity of the shimmer effect
Bloom	An elastic space reverb effect that uses algorithmic processing to simulate the propagation and reflection of sound in large, open spaces, enhancing the sense of spatial depth. When emulating a concert hall, it precisely calculates the paths and intensities of multiple sound reflections, layering them to produce rich depth and dimensionality for a convincingly realistic live-sound experience.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Tone: Adjusts the brightness of the wet reverb signal Lend: Adjusts the sustain and extension of the wet reverb signal Length: Controls the proportion of the wet reverb signal's duration within the overall audio
Cloud	A cloud reverb effect designed to deliver a lush, atmospheric ambience. The core sonic character evokes sound naturally convolving and diffusing through soft, billowing clouds — a single note can trigger multi-directional reflections that build a boundless, open soundstage. It preserves the clarity and transparency of the source while using dense spatial reflections to craft a grand, ethereal, and richly layered immersive listening experience.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Diff: Adjusts the smoothness and cohesion of reflections within the reverb

Effect Name	Feature Description	Parameter Adjustment
Lofi	A lo-fi reverb effect that merges lo-fi aesthetics with reverb processing to simulate the sonic character of vintage audio equipment, delivering a rich retro and nostalgic atmosphere.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Sample: Adjusts the number of times the audio signal is sampled per second Noise: Adjusts the level of noise present in the audio signal Mod: Adjusts the influence of the modulation signal on the carrier signal
Swell	A swell reverb effect featuring dynamic, progressive reverb that adjusts parameters in real time in response to the audio signal, producing vibrant and emotionally expressive sonic transformations that stand apart from the fixed character of conventional static reverb.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Tone: Adjusts the brightness of the wet reverb signal Lend: Adjusts the sustain and extension of the wet reverb signal Rise: Adjusts the rate at which the reverb builds up
Room stereo	A stereo room reverb effect that simulates the acoustic characteristics of a simple room: soft interior materials absorb a portion of the reflected sound, while the core signal reflects off the walls to form a natural reverb, faithfully reproducing the authentic sonic texture of a small space. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Mod: Adjusts the influence of the modulation signal on the carrier signal
Hall stereo	A stereo hall reverb effect that simulates the acoustic characteristics of large indoor spaces such as concert halls. The core sonic impression is one of expansive spatial depth with subtle diffusion — preserving detail clarity while naturally enveloping the dry signal to recreate a grand soundstage. Ideal for enhancing vocal transparency, extending the spatial presence of instruments (such as piano and strings), or imparting an authentic sense of "standing in a vast hall" to ambient sound and narration. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Mod: Adjusts the influence of the modulation signal on the carrier signal
Plate stereo	A stereo plate reverb effect that simulates the reflection of sound off a metal plate surface. As the audio signal reaches the plate, a portion is absorbed, a portion passes through, and the primary component reflects off the surface, generating the characteristic plate reverb response. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Mod: Adjusts the influence of the modulation signal on the carrier signal
Spring stereo	A stereo spring reverb effect representing one of the most iconic vintage reverb types in audio processing, with a distinctly mechanical character: the audio signal is transmitted into a spring assembly to induce resonance, and a dedicated transducer captures the harmonic feedback generated by the spring's vibration, ultimately simulating the sensation of being enclosed within a space fitted with spring-based acoustic structures. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Combs: Adjusts the intensity of the comb filter effect

Effect Name	Feature Description	Parameter Adjustment
Shimmer stereo	A stereo shimmer reverb effect in which the wet reverb signal incorporates pitch shifting, enabling creative atmospheric sound design — dissonant intervals evoke an eerie, unsettling backdrop, while harmonic intervals produce an ethereal, luminous quality, with reverb tails that bloom like the gentle glow of a rising dawn. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Tone: Adjusts the brightness of the wet reverb signal Pitch: Adjusts the pitch shift of the wet reverb signal Amount: Adjusts the intensity of the shimmer effect
Bloom stereo	A stereo elastic space reverb effect that uses algorithmic processing to simulate the propagation and reflection of sound in large, open spaces, enhancing the sense of spatial depth. When emulating a concert hall, it precisely calculates the paths and intensities of multiple sound reflections, layering them to produce rich depth and dimensionality for a convincingly realistic live-sound experience. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Tone: Adjusts the brightness of the wet reverb signal Lend: Adjusts the sustain and extension of the wet reverb signal Length: Controls the proportion of the wet reverb signal's duration within the overall audio
Cloud stereo	A stereo cloud reverb effect designed to deliver a lush, atmospheric ambience. The core sonic character evokes sound naturally convolving and diffusing through soft, billowing clouds — a single note can trigger multi-directional reflections that build a boundless, open soundstage. It preserves the clarity and transparency of the source while using dense spatial reflections to craft a grand, ethereal, and richly layered immersive listening experience. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal HPass: Adjusts the degree of low-frequency filtering applied to the wet signal LPass: Adjusts the degree of high-frequency filtering applied to the wet signal Diff: Adjusts the smoothness and cohesion of reflections within the reverb
Lofi stereo	A stereo lo-fi reverb effect that merges lo-fi aesthetics with reverb processing to simulate the sonic character of vintage audio equipment, delivering a rich retro and nostalgic atmosphere. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Sample: Adjusts the number of times the audio signal is sampled per second Noise: Adjusts the level of noise present in the audio signal Mod: Adjusts the influence of the modulation signal on the carrier signal
Swell stereo	A stereo swell reverb effect featuring dynamic, progressive reverb that adjusts parameters in real time in response to the audio signal, producing vibrant and emotionally expressive sonic transformations that stand apart from the fixed character of conventional static reverb. By simulating the differences in reflected sound received by the left and right ears in a real space, it constructs a spatially oriented stereo sound field.	Decay: Adjusts the duration of the reverb tail Mix: Adjusts the ratio of wet reverb signal to dry signal Tone: Adjusts the brightness of the wet reverb signal Lend: Adjusts the sustain and extension of the wet reverb signal Rise: Adjusts the rate at which the reverb builds up

The manufacturer and product names mentioned in this manual are used solely for the objective description of the audio characteristics and functional features of the relevant products. This does not constitute any recommendation, endorsement, or sponsorship of any manufacturer or product. All trademark names are the lawful property of their respective companies, and the use of these names in this manual does not infringe upon their trademark rights.

TANK PRO

音色效果列表

DRUM (100)

效果名称	功能描述
DRUM (100)	搭载 100 种丰富节奏类型，全面覆盖流行、摇滚、爵士、民谣、拉丁等多元音乐风格，满足不同场景的演奏与创作需求；支持灵活节奏加花功能，可通过即时加花设计增强旋律层次感与现场表现力，适配即兴演奏、编曲制作等多种使用场景。

WAH (14)

效果名称	功能描述	参数调节
X-Wah	踏板哇音效果，声音基于VOX Clyde McCoy哇音效果器音色。	Value: 调节哇音核心频率与频率范围的综合值 Gain: 调节哇音效果增益量 Level: 调节哇音效果音量
Funk-Wah	踏板哇音效果，声音基于Xotic XW - 2哇音效果器音色。	
Slide-Wah	踏板哇音效果，声音基于VOX V846哇音效果器音色。	
Cry-Wah	踏板哇音效果，声音基于Dunlop Cry Baby哇音效果器音色。	
Wah-Wah	"自动哇音效果，通过设定的频率参数进行有规律的波纹运动音色。"	Speed: 调节哇音效果速度 Q: 调节中心频率和波形宽度的比例 Width: 调节波形的宽度综合参数比例 Mix: 调节波形的宽度综合参数比例 Level: 调节哇音效果音量 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Sense-Wah	动态感应哇音效果，通过感应乐器的演奏动态自动触发并调节哇音效果。	Sense: 调节灵敏度 Attack: 调节触发效果后的启动时间 Q: 调节中心频率和波形宽度的比例 fPeak: 调节湿声里扫频效果的反馈程度 Mix: 调节波形的宽度综合参数比例 Width: 调节波形的宽度综合参数比例 Level: 调节哇音效果音量
Compress	压缩效果，控制声音动态。	Sustain: 调节综合调压压缩阈值和比例 Attack: 调节压缩启动速度 Level: 调节压缩后的音量 Blend: 调节压缩声与原始声的混合比例
Compress Pro	专业压缩效果，可调节多个参数。	Ratio: 调节压缩比 Gain: 调节输入增益 Knee: 调节压缩启动的过渡平滑度 Thd: 调节压缩触发的阈值 Attack: 调节压缩启动速度 Level: 调节压缩后的音量 Blend: 调节压缩声与原始声的混合比例
F Compress	压缩效果，兼具动态控制与可调节压缩后的明亮度。	Ratio: 调节压缩比 Gain: 调节输入增益 Knee: 调节压缩启动的过渡平滑度 Thd: 调节压缩触发的阈值 Attack: 调节压缩启动速度 Tone: 调节压缩效果的明亮度 Level: 调节压缩后的音量 Blend: 调节压缩声与原始声的混合比例
Boost	激励效果，核心是提升信号强度。	Gain: 调节输入增益 Level: 调节经激励效果后的音量
A Boost	A型激励效果。	Gain: 调节输入增益
B Boost	B型激励效果。	Bass: 调节激励效果的低频
E Boost	E型激励效果。	Mid: 调节激励效果的中频 Treble: 调节激励效果的高频 Level: 调节经激励效果后的音量
Boost ED	带过载功能的过载型激励效果。	Gain: 调节输入增益 Grit: 调节激励效果的过载饱和度 Level: 调节经激励效果后的音量

FX1&FX2 (20)

效果名称	功能描述	参数调节
Noise Gate++	降噪效果，AI 驱动的信号处理工具，控制信号通断。	Gate: 调节噪音门触发阈值
AI Gate MS	降噪效果，基于统计偏差跟进模型的智能算法。	Gate: 调节噪音门触发阈值 Bias: 调节降噪预准备的提前量 Thd: 调节降噪阈值
Soft Gate	降噪效果，监测并控制音频电平，实现降噪、提清清晰度与音质。	Thd: 调节降噪阈值
Hard Gate	降噪效果，处理音频信号时，比 Soft Gate 更激进。	Att: 调节信号超阈后降噪门全开时间 Rel: 调节信号低阈后降噪门的关闭时间 Thd: 调节降噪阈值 Kw: 调节阈值上下过渡区宽度 Ratio: 调节输入噪声与输出信号比例
Pro Gate	降噪效果，设阈值精准识别并去除低阈值背景噪，优化音频纯净度。	Sustain: 调节综合调压缩阈值和比例 Attack: 调节压缩启动速度 Level: 调节压缩后的音量 Blend: 调节压缩声与原始声的混合比例
Compress	压缩效果，控制声音动态。	Ratio: 调节压缩比 Gain: 调节输入增益 Knee: 调节压缩启动的过渡平滑度 Thd: 调节压缩触发的阈值 Attack: 调节压缩启动速度 Level: 调节压缩后的音量 Blend: 调节压缩声与原始声的混合比例
Compress Pro	专业压缩效果，可调节多个参数。	Ratio: 调节压缩比 Gain: 调节输入增益 Knee: 调节压缩启动的过渡平滑度 Thd: 调节压缩触发的阈值 Attack: 调节压缩启动速度 Level: 调节压缩后的音量 Blend: 调节压缩声与原始声的混合比例
F Compress	压缩效果，兼具动态控制与可调节压缩后的明亮度。	Ratio: 调节压缩比 Gain: 调节输入增益 Knee: 调节压缩启动的过渡平滑度 Thd: 调节压缩触发的阈值 Attack: 调节压缩启动速度 Tone: 调节压缩效果的明亮度 Level: 调节压缩后的音量 Blend: 调节压缩声与原始声的混合比例
Boost	激励效果，核心是提升信号强度。	Gain: 调节输入增益 Level: 调节经激励效果后的音量
A Boost	A型激励效果。	Gain: 调节输入增益
B Boost	B型激励效果。	Bass: 调节激励效果的低频
E Boost	E型激励效果。	Mid: 调节激励效果的中频 Treble: 调节激励效果的高频 Level: 调节经激励效果后的音量
Boost ED	带过载功能的过载型激励效果。	Gain: 调节输入增益 Grit: 调节激励效果的过载饱和度 Level: 调节经激励效果后的音量
Lofi	低保真效果，模拟老式播放设备的音质特点。	Bit: 调节比特率 Level: 调节效果音量 filter: 调节滤波器的特定频段
Pitch	音高变化效果，通过改变音频信号的频率来实现音高的变化。	HPitch: 调节高音部分的音高 LPitch: 调节低音部分的音高 HighLvl: 调节高音部分的音量 LowLvl: 调节低音部分的音量 DryLvl: 调节原始干声的音量
Pitch shifter	正负12个半音移调效果。	Semi: 音调偏移量
Octave	八度效果，能给声音添加“高八度”或“低八度”的效果。	HighLvl: 调节“高八度声音”的音量 LowLvl: 调节“低八度声音”的音量 DryLvl: 调节原始干声的音量
Ring	模拟铃声效果，模拟了一种类似钟声、铃声的回响效果。	Freq: 调节铃声效果的频率 Mix: 调节铃声效果与原始声音的混合比例
Exp_Vol	外接表情踏板器的音量控制效果，建议配合Pedal设置中Toe的None档位使用。	Vol: 调节音量（请将Vol参数映射到表情踏板上使用）
Exp_Pan	外接表情踏板器的左右声道输出比例控制效果，建议配合Pedal设置中Toe的Resume档位使用。	Pan: 调节左右声道输出比例（请将Pan参数映射到表情踏板上使用）

DS (30)

命名	原型	参数调节
另附音色列表	另附音色列表	Gain: 调节模块的增益度 Level: 调节模块的音量 Bass: 调节模块的低频 Middle: 调节模块的中频 Treble: 调节模块的高频 Reso: 调节模块低频的共鸣度 Pres: 调节模块的临场度 Bright: 调节模块的明亮度

AMP (90)

命名	原型	参数调节
另附音色列表	另附音色列表	Gain: 调节模块的增益度 Level: 调节模块的音量 Bass: 调节模块的低频 Middle: 调节模块的中频 Treble: 调节模块的高频 Reso: 调节模块低频的共鸣度 Pres: 调节模块的临场度 Bright: 调节模块的明亮度

CAB (90)

命名	原型	参数调节
另附音色列表	另附音色列表	Level: 调节箱体的输出音量 Lcut: 调节箱体低切频点 Hcut: 调节箱体高切频点

EQ (8)

效果名称	功能描述	参数调节
Guitar EQ 6	吉他 6 段均衡器，能精细调节吉他不同频率的音色。	100Hz: 调节100HZ附近频率响应 200Hz: 调节200HZ附近频率响应 400Hz: 调节400HZ附近频率响应 800Hz: 调节800HZ附近频率响应 1.6kHz: 调节1600HZ附近频率响应 3.2kHz: 调节3200HZ附近频率响应
Bass EQ 7	贝斯 7 段均衡器，用于精细调节贝斯不同频率的音色。	50Hz: 调节50HZ附近频率响应 120Hz: 调节120HZ附近频率响应 400Hz: 调节400HZ附近频率响应 500Hz: 调节500HZ附近频率响应 800Hz: 调节800HZ附近频率响应 4.5kHz: 调节4500HZ附近频率响应 10kHz: 调节10000HZ附近频率响应
Normal EQ 10	10 段均衡器，可精细调节从超低频到超高频的声音。	31.25Hz: 调节31.5HZ附近频率响应 62.5Hz: 调节62.5HZ附近频率响应 125Hz: 调节125HZ附近频率响应 250Hz: 调节250HZ附近频率响应 500Hz: 调节500HZ附近频率响应 1kHz: 调节1000HZ附近频率响应 2kHz: 调节2000HZ附近频率响应 4kHz: 调节4000HZ附近频率响应 8kHz: 调节8000HZ附近频率响应 16kHz: 调节16000HZ附近频率响应
Lofi	低保真效果，模拟老式播放设备的音质特点。	Bit: 调节比特率 Level: 调节效果音量 filter: 调节滤波器的特定频段
Pitch	音高变化效果，通过改变音频信号的频率来实现音高的变化。	HPitch: 调节高音部分的音高 LPitch: 调节低音部分的音高 HighLvl: 调节高音部分的音量 LowLvl: 调节低音部分的音量 DryLvl: 调节原始干声的音量
Pitch shifter	正负12个半音移调效果。	Semi: 音调偏移量
Octave	八度效果，能给声音添加“高八度”或“低八度”的效果。	HighLvl: 调节“高八度声音”的音量 LowLvl: 调节“低八度声音”的音量 DryLvl: 调节原始干声的音量
Ring	模拟铃声效果，模拟了一种类似钟声、铃声的回响效果。	Freq: 调节铃声效果的频率 Mix: 调节铃声效果与原始声音的混合比例

效果名称	功能描述	参数调节
Chorus	合唱效果，模拟多人合唱或多乐器合奏的听感，让单一音色更丰满、有空间感和厚度，是提升音色表现力的常用工具。	Speed: 调节效果的速度 Depth: 调节效果的深度 Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Tri Chorus	三重合唱效果，普通合唱的进阶形式，通过生成3个独立副信号，模拟更自然丰满的多人/多乐器听感，比传统合唱层次感更突出、空间感更细腻。	Speed: 调节效果的速度 Depth: 调节效果的深度 Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Flanger	朗格效果，通过延迟信号与原始信号叠加，产生周期性“扫频”感（类似飞机掠过的涟漪声），赋予音色强烈动态和空间流动感，常用于电吉他、合成器等乐器。	Speed: 调节效果的速度 Depth: 调节效果的深度 Fb: 调节湿声中扫频效果的反馈量 Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Tri Flanger	三重佛朗格效果，普通 Flanger 的进阶形式，通过三个独立镶边信号叠加，生成更复杂、层次更细腻的扫频波动感，听感更丰富（类似多把镶边乐器同时发声），适合需强烈动态且避免单调的场景。	Speed: 调节效果的速度 Depth: 调节效果的深度 Fb: 调节湿声中扫频效果的反馈量 Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Tremolo	颤音效果，周期性改变音量大小（不改变音高，与揉弦 Vibrato 区分），产生呼吸感或脉冲感，类似快速手动调节音量的听感。	Speed: 调节效果的速度 Depth: 调节效果的深度 Level: 调节模块的音量 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Tri Tremolo	三重颤音效果，传统颤音的进阶版，以三个交织的节奏替代单一规律，打造更丰富有层次的音量波动听感，避免单调，提升音色动态复杂度。	Speed: 调节效果的速度 Depth: 调节效果的深度 Level: 调节模块的音量 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Opto Tremolo	光学颤音，近似老式 Fender 音箱颤音，更显棱角与清脆。颤音强度随触弦力度变化，动态清晰，对高频作用明显，带 Vibe 韵味，传统自然，可润色音色、营造独特意境。	Speed: 调节效果的速度 Depth: 调节效果的深度 Level: 调节模块的音量 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Phaser	相移效果，通过改变音频信号相位并与原始信号混合，产生类似“扫频”的空灵、迷幻音效，适用于吉他等乐器处理及影视音频制作。其梳状滤波器凹口非谐波相关，音色柔和、嘶嘶声温和，无镶边效果的尖锐金属感。	Speed: 调节效果的速度 MidCut: 调节中频区域相移作用强度 Reso: 调节特定频率范围的共振量 Fb: 调节湿声中扫频效果的反馈量 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Vibrato	震音效果，通过音高周期性波动产生效果，核心是“音高波动”，区别于颤音（Tremolo）的“音量波动”，广泛应用于乐器演奏和音频处理。本质是原始音高围绕基准音高规律性上下波动，类似人声演唱时的自然音高伏。	Speed: 调节效果的速度 Depth: 调节效果的深度 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Tri Vibrato	三重震音效果，通过三种不同速率、深度或波形的震音效果，对同一音频信号叠加或交错调制，提升音色的层次感与表现力（例如低速浅度营造轻柔韵律、中速中度增加动态、高速可调深度强化起伏）。	Speed: 调节效果的速度 Depth: 调节效果的深度 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Opto Vibrato	光学震音效果，基于光学原理调制音频信号振幅，产生自然柔和的周期性音量变化，失真小。能为声音增添动态感、让音色更饱满生动，常用于增强乐器表现力（尤其吉他），可营造复古、温暖的音效氛围。	Speed: 调节效果的速度 Depth: 调节效果的深度 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Univibe	多谐振荡器效果，最初旨在模拟旋转扬声器声音，实际创造出独特的“漩涡式”音色，兼具温暖、迷幻特质，是吉他手和音乐制作人的标志性工具。通过结合相位调制与动态滤波，音色更具“有机感”，带有明显低频谐波和“呼吸感”。	Speed: 调节效果的速度 Depth: 调节效果的深度 Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Tri Univibe	三重多谐振荡器效果，基于多谐振荡器原理，通过三个振荡器相互作用，产生复杂的相位与幅度调制，形成“漩涡式”听感，仿佛声音在空间中扩散收缩，兼具强烈的空间感和动态感。	Speed: 调节效果的速度 Depth: 调节效果的深度 Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致

效果名称	功能描述	参数调节
Autofilter	自动滤波效果，动态音频效果器，通过预设或外部触发信号自动控制滤波器参数，实现音色动态变化，突破手动调节局限。广泛应用于吉他、合成器及电子音乐制作，是塑造“动感音色”的关键工具，能连接演奏表情、编曲律动与音色设计，让声音更具生命力。	Speed: 调节效果的速度 Min: 扫频频率最小值 Max: 扫频频率最大值 Mix: 调节效果湿声与干声的比例 Fb: 调节湿声中扫频效果的反馈量 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Phaser Stereo	立体声相移效果，基于传统单声道相移效果升级，保留相位调制扫频核心质感，通过左右声道差异化处理，赋予音效更强的空间纵深感与动态运动感，是塑造“流动感”“空间穿梭感”的关键工具。其核心优势在于左右声道相位变化不同步，使扫频效果从平面音色变化升级为立体空间运动。	Speed: 调节效果的速度 Stage: 选择频点处理模式。A 为单个频率点相移；B 为多个频率点同时相移。 Regen: 调节效果的反馈强度 Mix: 调节效果湿声与干声的比例 Lfo: 选择低频振荡器波形，包括正弦波 (Sin)、方形波 (Sqw)、锯齿波 (Saw)、三角波 (Tri) 和特制波形 (Wav) Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Flanger Stereo	立体声弗朗格效果，基于传统单声道弗朗格效果升级，保留“延迟叠加扫频”的核心质感（如“飞机掠过”的金属味或“磁带高速擦过”的丝滑感），通过左右声道差异化处理，让音效从平面波动升级为立体空间运动，是塑造“环绕感”“动态穿梭感”的关键工具。	Speed: 调节效果的速度 Depth: 调节效果的深度 Regen: 调节效果的反馈强度 Mix: 调节效果湿声与干声的比例 Jet: 调节效果喷射感的强度 Lfo: 选择低频振荡器波形，包括正弦波 (Sin)、方形波 (Sqw)、锯齿波 (Saw)、三角波 (Tri) 和特制波形 (Wav) Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Vibe Stereo	立体声多谐振荡器效果，借助深度正反馈使电子器件交替导通截止，产生含丰富高次谐波的矩形脉冲波，通过左右声道差异化调制参数营造立体声效果。与立体声弗朗格效果（延迟叠加扫频，有“喷气声”）、立体声相移效果（全通波调制频率峰值）不同，它侧重创造旋转、波动的空间感及独特谐波音色，还能呈现合唱等多样化效果。	Speed: 调节效果的速度 Stage: 选择频点处理模式。A 为单个频率点相移；B 为多个频率点同时相移。 Regen: 调节效果的反馈强度 Mix: 调节效果湿声与干声的比例 Lfo: 选择低频振荡器波形，包括正弦波 (Sin)、方形波 (Sqw)、锯齿波 (Saw)、三角波 (Tri) 和特制波形 (Wav) Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Chorus Stereo	立体声合唱效果，广泛应用的调制类音频效果，核心是模拟“多人合唱”或“多乐器同步演奏”的自然差异，在立体声场中打造丰富的空间层次与音色厚度，既能拓宽声音宽度，又能赋予自然动感，是塑造乐器、人声空间感的重要工具。工作原理是将处理后的“分身信号”分配到左右声道，通过调节相位、延迟差或音量比例制造声道差异，再与原始信号混合，形成声音在立体空间中扩散的听感。	Speed: 调节效果的速度 Depth: 调节效果的深度 Mode: 选择调制模型，包括特制谐波 (Harm)、正弦波 (Sin)、方形波 (Sqw)、锯齿波 (Saw)、三角波 (Tri) 和特制波形 (Wav) Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Tremolo Stereo	立体声颤音效果，通过左右声道差异化的音量周期性波动塑造空间感的调制效果，能增强声音律动感、打破单调，是塑造乐器与人声“呼吸感”的常用工具。左右声道的差异波动可产生多样听感，如左右强弱时的“声音左右移动”，相位相反时的“左右交替跳动”，让声音摆脱固定位置限制。	Speed: 调节效果的速度 Depth: 调节效果的深度 Level: 调节模块的音量 Mode: 选择调制模型，包括正弦波 (Sin)、带失真的电流 (Tube)、残影模拟 (Phntn) Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致
Vibrato Stereo	立体声震音效果，核心是通过音高周期性波动丰富音色（区别于颤音的音量波动），并经左右声道差异化处理，让声音从“平面”升级为“立体”，营造更强的空间感与层次感。其特色是“音高波动的空间化”，不强调厚度或音量变化，是提升音色表现力的实用工具。	Speed: 调节效果的速度 Depth: 调节效果的深度 Bits: 调节数字信号处理的量化精度 Mix: 调节效果湿声与干声的比例 Sync: 开启后，Speed 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM保持一致

效果名称	功能描述	参数调节
Clean	纯净延迟效果，生成自然简洁的回声，无复杂调制，保留原始音色纯净，适配简约真实的音色需求。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Modern	现代延迟效果，对延迟信号做相移处理，结合 LFO 调制，让回声兼具空间纵深与动态层次，避免单调，听感丰富有生命力，同时保留音色基底的清晰度。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Phaser: 调节延迟信号的相位调制强度 Mod: 调节低频振荡器（LFO）对全通滤波器的调制强度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Echo	回音延迟效果，模拟声音在空间中的反射与衰减，通过回声的重复和音量递减，赋予声音空间感、深度感或独特节奏层次，还原声波传播的物理特性	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Analog	模拟延迟效果，用 DSP 技术复刻传统 BBD 芯片的音色与电路特性，兼顾数字设备的稳定性，还原模拟延迟温暖湿润、富有生命力的经典质感，重点复现高频自然衰减和偶次谐波失真的独特效果。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Duck	闪避延迟效果，动态监测干信号电平，干音发声时延迟湿音自动减弱（避免冲突），干音减弱 / 停顿时湿音自然增强；Release 参数决定湿音恢复速度，短则突兀，长则平滑。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Release: 调节延迟信号音量恢复正常的速度 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Dtype	磁带延迟效果，叠加磁带模拟处理，赋予回声老式磁带的复古质感与瑕疵感；核心参数 Grit 控制处理强度（值越大声音越模糊，为 0 时仅LFO 作用于延迟信号），同时通过 LFO 实现音高周期性轻微波动。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Grit: 调节磁带模拟效果对延迟信号的处理强度 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Tremolo	颤音延迟效果，给延迟信号叠加周期性音量起伏的颤音，让回声既有时间间隔，又带动态音量波动，增强律动感与空间层次。颤音周期 = Time × Pattern，Pattern 值越小起伏越密集，等于 1/1 时与 Time 同步，越大起伏越舒展。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Pattern: 调节颤音一个周期的时值 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Filter	滤波延迟效果，给延迟信号叠加滤波处理，让每次回声音色产生差异，打造更复杂的空间感与音色层次。核心参数 Filter 控制滤波强度，值越大处理越明显，值为 0 时无滤波效果。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Filter: 调节滤波效果的强度 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Dual	双延迟效果，通过 T-Mode 参数调节两个延迟信号的频率差（时间差 = Time×T-Mode），产生双重回声、空间叠加等复杂听感。T-Mode 小于1/1 时，回声弱→强循环；等于 1/1时，主次回声重叠循环；大于 1/1时，回声强→弱循环。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 T-Mode: 调节主次延迟信号的时值 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Lofi	低保真延迟效果，结合延迟处理与复古失真特性，通过 Grit 参数添加饱和失真和背景噪声，搭配 LFO 调制模拟设备不稳定波动，营造怀旧粗糙的音频质感，适配复古音乐氛围。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Grit: 调节复古失真效果对延迟信号的处理强度 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致

效果名称	功能描述	参数调节
Pattern	模式延迟效果，以预设节奏型为核心，延迟回声按预设的节奏 Pattern（如特定拍型、鼓点逻辑）重复，而非固定时间间隔；搭配 LFO 调制（Speed 控速度、Depth 控幅度）为节奏框架加入周期性变化，避免机械感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Pattern: 调节回声的节奏模式（共 15 种） Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Ice	冰延迟效果，对延迟信号进行音高调整，搭配 LFO 动态调制，让回声产生独特和声织体与层次感，从单纯复刻升级为参与音乐结构的创作元素，兼具空间纵深感与情感张力	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Pitch: 调节延迟回声的音高 Mod: 调节低频振荡器（LFO）的调制强度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Reverse	反向延迟效果，对延迟信号进行时间反向处理，再经移相器调整相位，产生声音从“未来”倒回“现在”的独特听感，兼具新奇感与神秘感，能增强声音的空间感和立体感	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Phaser: 调节移相处理后的相位差信号与原始反向信号的混合比 Mod: 调节移相器中低频振荡器（LFO）的调制强度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
PingPong Stereo	立体声乒乓延迟效果，左右声道交替触发延迟信号，模拟声音“左右弹跳”的动态轨迹，利用立体声相位差与时间差打造强烈空间流动感；区别于普通立体声延迟的“独立回声”，其信号存在因果关系，流动性更突出。立体声纯净延迟效果，生成自然简洁的回	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Clean Stereo	声，无复杂调制，保留原始音色纯净，适配简约真实的音色需求。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Modern Stereo	立体声现代延迟效果，对延迟信号做相移处理，结合 LFO 调制，让回声兼具空间纵深与动态层次，避免单调，听感丰富有生命力，同时保留音色基底的清晰度。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Phaser: 调节延迟信号的相位调制强度 Mod: 调节低频振荡器（LFO）对全通滤波器的调制强度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Echo Stereo	立体声回声延迟效果，模拟声音在空间中的反射与衰减，通过回声的重复和音量递减，赋予声音空间感、深度感或独特节奏层次，还原声波传播的物理特性。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Analog Stereo	立体声模拟延迟效果，用 DSP 技术复刻传统 BBD 芯片的音色与电路特性，兼顾数字设备的稳定性，还原模拟延迟温暖湿润、富有生命力的经典质感，重点复现高频自然衰减和偶次谐波失真的独特效果。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Duck Stereo	立体声闪避延迟效果，动态监测干信号电平，干音发声时延迟湿音自动减弱（避免冲突），干音减弱 / 停顿时湿音自然增强；Release 参数决定湿音恢复速度，短则突兀，长则平滑。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Release: 调节延迟信号音量恢复正常的速度 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Dtype Stereo	立体声磁带延迟效果，叠加磁带模拟处理，赋予回声老式磁带的复古质感与瑕疵感；核心参数 Grit 控制处理强度（值越大声音越模糊，为 0 时仅 LFO 作用于延迟信号），同时通过 LFO 实现音高周期性轻微波动。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Grit: 调节磁带模拟效果对延迟信号的处理强度 Speed: 调节低频振荡器（LFO）的变化速度 Depth: 调节低频振荡器（LFO）的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致

效果名称	功能描述	参数调节
Tremolo Stereo	立体声颤音延迟效果，给延迟信号叠加周期性音量起伏的颤音，让回声既有时间间隔，又带动态音量波动，增强律动感与空间层次。颤音周期 = Time × Pattern，Pattern 值越小起伏越密集，等于 1/1 时与 Time 同步，越大起伏越舒展。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Pattern: 调节颤音一个周期的时值，影响起伏密集度 Speed: 调节颤音的变化速度 Depth: 调节颤音的变化深度 Dive: 调节颤音的下潜深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Filter Stereo	立体声滤波延迟效果，给延迟信号叠加滤波处理，让每次回声音色产生差异，打造更复杂的空间感与音色层次。核心参数 Filter 控制滤波强度，值越大处理越明显，值为 0 时无滤波效果。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Filter: 调节滤波效果的强度 Speed: 调节颤音的变化速度 Depth: 调节颤音的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Dual Stereo	立体声双延迟效果，通过 T-Mode 参数调节两个延迟信号的频率差（时间差 = Time×T-Mode），产生双重回声、空间叠加等复杂听感。T-Mode小于 1/1 时，回声弱→强循环；等于1/1 时，主次回声重叠循环；大于1/1 时，回声强→弱循环。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 T-Mode: 调节主次延迟信号的时值 Speed: 调节颤音的变化速度 Depth: 调节颤音的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Lofi Stereo	立体声低保真延迟效果，结合延迟处理与复古失真特性，通过 Grit 参数添加饱和失真和背景噪声，搭配 LFO调制模拟设备不稳定波动，营造怀旧粗糙的音频质感，适配复古音乐氛围。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Grit: 调节复古失真效果对延迟信号的处理强度 Speed: 调节颤音的变化速度 Depth: 调节颤音的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Pattern Stereo	立体声模式延迟效果，以预设节奏型为核心，延迟回声按预设的节奏 Pattern（如特定拍型、鼓点逻辑）重复，而非固定时间间隔；搭配 LFO 调制（Speed 控速度、Depth 控幅度）为节奏框架加入周期性变化，避免机械感。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Pattern: 调节回声的节奏模式（共 15 种） Speed: 调节颤音的变化速度 Depth: 调节颤音的变化深度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Ice Stereo	立体声冰延迟效果，对延迟信号进行音高调整，搭配 LFO 动态调制，让回声产生独特和声织体与层次感，从单纯复刻升级为参与音乐结构的创作元素，兼具空间纵深感与情感张力。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Pitch: 调节延迟回声的音高 Mod: 调节低频振荡器（LFO）的调制强度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致
Reverse Stereo	立体声反向延迟效果，对延迟信号进行时间反向处理，再经移相器调整相位，产生声音从“未来”倒回“现在”的独特听感，兼具新奇感与神秘感，能增强声音的空间感和立体感。利用声道间的时间差与空间差异，增强声音的宽度感和包围感，营造出更开阔的立体空间听感。	Time: 调节原始声与延迟声的时间间隔 Fb: 调节延迟回声的重复次数 Mix: 调节延迟湿声与干声的比例 Phaser: 调节移相处理后的相位差信号与原始反向信号的混合比例 Mod: 调节移相器中低频振荡器（LFO）的调制强度 Sync: 开启后，Time 按拍数显示（同步预设BPM） S-Bpm: 同步速度，与预设BPM 保持一致

效果名称	功能描述	参数调节
Room	房间混响效果，模拟简易房间的声学特性：室内软质材料会吸收部分反射声，核心声音通过墙壁反射形成自然混响，还原小型空间的真实听觉质感。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Depth: 调节早期反射与晚期混响的比例
Hall	大厅混响效果，是模拟音乐厅等大型室内声学特性的音频效果，核心听感为开阔空间纵深感与轻微散射感——不糊细节且能自然包裹原声，还原宏大声场；适合增强人声通透感、乐器（如钢琴、弦乐）空间延展性，或给环境音、旁白等赋予“置身空旷大厅”的真实沉浸感。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Depth: 调节早期反射与晚期混响的比例
Plate	金属板反射混响效果，模拟声音在金属板上的反射过程。当声音信号到达金属板表面时，一部分声音会被吸收，一部分会穿透金属板，而主要部分则会在金属板表面发生反射。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Mod: 调节调制信号对载波信号的影响程度
Spring	弹簧混响效果，是音频处理中极具标志性的复古混响类型，其工作原理极具机械质感：通过将音频信号传输至弹簧组件并激发共振，再由专用拾音器捕捉弹簧振动产生的谐波反馈，最终模拟出“仿佛置身于装有弹簧声学结构的密闭空间”的混响效果。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Combs: 调节梳状滤波效应的强度
Shimmer	微光混响效果，混响湿音带音高偏移，能创意性营造氛围——调不和谐音程可做恐怖背景声，调和谐音程则出空灵“美颜感”，混响尾音像“日出微光渐亮”般美妙。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Tone: 调节混响湿声的明暗度 Pitch: 调节混响湿声的音高偏移 amount: 调节微光效果的强弱程度
Bloom	弹性空间混响效果，通过算法模拟声音在大型开阔空间的传播与反射，增强空间感；模拟音乐厅时，会精确计算声音多次反射的路径与强度，叠加上丰富层次和立体感，营造逼真现场感。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Tone: 调节混响湿声的明暗度 Lend: 调节混响湿声的延展持续度 Length: 控制混响湿声在整个音频里的时长占
Cloud	云混响效果，是主打“华丽氛围感”的大型环境混响，核心听感如同声音从蓬松云层中自然卷积扩散——单个音符即可触发多方向的反射声，构建出无边界的开阔声场，既保留声音的通透质感，又能通过密集的空间反射，营造出宏大、空灵且富有层次的沉浸式听觉体验。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Diff: 调节混响里反射声的融合细腻度
Lofi	低保真混响效果，融合低保真与混响，模拟老式音频设备质感，主打浓郁复古怀旧氛围。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Sample: 调节每秒音频信号的采样次数 Noise: 调节音频中噪声的强弱 Mod: 调节调制信号对载波信号的影响程度
Swell	膨胀混响效果，动态渐进式混响，能随音频信号实时调整参数，带来富有活力与情感的动态声音变化，区别于传统静态混响的固定听感。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Tone: 调节混响湿声的明暗度 Lend: 调节混响湿声的延展持续度 Rise: 调节混响升强速率
Room stereo	立体声房间混响效果，模拟简易房间的声学特性：室内软质材料会吸收部分反射声，核心声音通过墙壁反射形成自然混响，还原小型空间的真实听觉质感。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Mod: 调节调制信号对载波信号的影响程度
Hall stereo	立体声大厅混响效果，是模拟音乐厅等大型室内声学特性的音频效果，核心听感为开阔空间纵深感与轻微散射感——不糊细节且能自然包裹原声，还原宏大声场；适合增强人声通透感、乐器（如钢琴、弦乐）空间延展性，或给环境音、旁白等赋予“置身空旷大厅”的真实沉浸感。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Mod: 调节调制信号对载波信号的影响程度
Plate stereo	立体声金属板反射混响效果，模拟声音在金属板上的反射过程。当声音信号到达金属板表面时，一部分声音会被吸收，一部分会穿透金属板，而主要部分则会在金属板表面发生反射。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Mod: 调节调制信号对载波信号的影响程度
Spring stereo	立体声弹簧混响效果，是音频处理中极具标志性的复古混响类型，其工作原理极具机械质感：通过将音频信号传输至弹簧组件并激发共振，再由专用拾音器捕捉弹簧振动产生的谐波反馈，最终模拟出“仿佛置身于装有弹簧声学结构的密闭空间”的混响效果。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Combs: 调节梳状滤波效应的强度

效果名称	功能描述	参数调节
Shimmer stereo	立体声微光混响效果，混响湿音带音高偏移，能创意性营造氛围——调不和谐音程可做恐怖背景声，调和谐音程则出空灵“美颜感”，混响尾音像“日出微光渐亮”般美妙。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Tone: 调节混响湿声的明暗度 Pitch: 调节混响湿声的音高偏移 amount: 调节微光效果的强弱程度
Bloom stereo	立体声弹性空间混响效果，通过算法模拟声音在大型开阔空间的传播与反射，增强空间感；模拟音乐厅时，会精确计算声音多次反射的路径与强度，叠加出丰富层次和立体感，营造逼真现场感。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Tone: 调节混响湿声的明暗度 Lend: 调节混响湿声的延展持续度 Length: 控制混响湿声在整个音频里的时长占比
Cloud stereo	立体声云混响效果，是主打“华丽氛围感”的大型环境混响，核心听感如同声音从蓬松云层中自然卷积扩散——单个音符即可触发多方向的反射声，构建出无边界的开阔声场，既保留声音的通透质感，又能通过密集的空间反射，营造出宏大、空灵且富有层次的沉浸式听觉体验。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 HPass: 调节湿声中低频成分的过滤程度 LPass: 调节湿声中高频成分的过滤程度 Diff: 调节混响里反射声的融合细腻度
Lofi stereo	立体声低保真混响效果，融合低保真与混响，模拟老式音频设备质感，主打浓郁复古怀旧氛围。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Sample: 调节每秒音频信号的采样次数 Noise: 调节音频中噪声的强弱 Mod: 调节调制信号对载波信号的影响程度
Swell stereo	立体声膨胀混响效果，动态渐进式混响，能随音频信号实时调整参数，带来富有活力与情感的动态声音变化，区别于传统静态混响的固定听感。通过模拟真实空间中左右耳接收到的反射声差异，构建出具有方位感的立体声场。	Decay: 调节混响尾音的持续时间 Mix: 调节混响湿声与干声的比例 Tone: 调节混响湿声的明暗度 Lend: 调节混响湿声的延展持续度 Rise: 调节混响升强速率

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